

FreshLens

AI-Powered Freshness Monitoring System
for Small-Scale Retailers

Test Plan Report

Execution results and screenshot evidence
27 September 2026

Unit · API · database security · service integration
Playwright browser UI · Grafana k6 performance

Contents

1. Evaluation mission and results.....	3
2. Scope, dependencies and environment	4
3. Test layering and techniques	5
4. Detailed functional test cases.....	6
5. INT-01 — Live service integration.....	7
6. Performance testing with Grafana k6	8
7. UI-02 — Browser testing with Playwright.....	10
8. Evidence, findings and reproducibility	11
9. Implemented inventory and references.....	12
Appendix A. Screenshot evidence	13

1. Evaluation mission and results

FreshLens is an AI-powered freshness monitoring system for small-scale grocery retailers. This report records the test plan, execution results and screenshot evidence for **CS3203, Group 21, PID 5**, on **27 September 2026**. The system combines a FastAPI service, PostgreSQL tenant isolation, Redis/Celery background processing, a Next.js administration interface and an Expo mobile application.

The evaluation targets the highest-impact risks: cross-tenant data access, duplicate or excessive stock deductions, failed scan orchestration, incorrect confidence-gate logic and unusable authentication screens. Functional correctness, browser behaviour and performance are reported separately.

1.1 Execution summary

Test group	Type	Recorded outcome	Evidence
API suite	Unit and API/component functional	68 passed ; 1 dependency warning	E07; detail E01–E02
ML suite	Unit and worker/data utility	28 passed	E08; detail E04
Web utilities	Frontend unit	8 passed	E05
Mobile utilities	Frontend unit	9 passed	E06
PostgreSQL security	Database integration/security	2 SQL suites passed	E03
Live service integration	HTTP, database and queue integration	14 passed in 2.49 s	E10
Playwright UI	Browser functional smoke	8 passed in 12.79 s	E20–E21; JUnit/log
Static analysis	ESLint and TypeScript	3 commands passed	E09
k6 smoke / load	Local API performance	PASS / PASS	E11–E16
k6 spike	Local API performance	FAIL: read and write p95	E17–E19

There are **135 passing automated test cases** across the API, ML, web, mobile, integration and Playwright suites. SQL suites, static checks and k6 assertions are reported separately. The selected API-01, API-02 and ML-01 screenshots show subsets of the full suites and are not added again to the total.

Evaluation outcome: the recorded functional cases pass. The short 25-user spike fails the proposed local latency criteria despite zero HTTP failures and correct stock balances. Performance acceptance is therefore incomplete. These results support the tested local scope; they do not establish production readiness or complete requirements coverage.

1.2 Revision and provenance

The terminal captures identify branch `test-plan-report` and base commit `2d9e4b1`. The later integration and performance executions include an **uncommitted sales concurrency fix** and newly added test files. The commit shown in their headers is a base revision, not a claim that the working tree was clean. Playwright was run separately during report completion against the same local checkout. All dates and screenshot filenames use Sri Lanka time (UTC+05:30), unless an artifact explicitly records UTC.

2. Scope, dependencies and environment

2.1 Target items and risks

Target	Tested behaviour	Main risk addressed
FastAPI scans and sales	Validation, roles, asynchronous acceptance, sale replay and rollback	Invalid requests or duplicate stock deduction
PostgreSQL	RLS, restricted login, tenant-scoped reads/writes, transaction integrity	Data leakage and inconsistent inventory
Redis/Celery pipeline	Real scan delivery, worker completion and result persistence	Scans accepted but never completed
ML decision utilities	Label mapping, confidence gates, dataset and batch utilities	Incorrect handling of uncertain predictions
Web and mobile utilities	Claims, aggregate mapping and session storage	Incorrect role/tenant handling or corrupt sessions
Next.js browser UI	Login controls, validation, redirects and narrow layout	Inaccessible or broken authentication flow
Local API under load	Read/write latency, HTTP failures and post-run stock	Degradation during concurrent activity

2.2 Execution environment

Item	Configuration used for this evidence
Host	Linux development laptop; shared with other local applications
Python / test runner	Python 3.12.3; pytest 8.4.1; <code>.venv - test - report t</code>
JavaScript	Node.js 22.23.2; <code>node: test through tsx</code>
Live integration stack	One Uvicorn worker; PostgreSQL 16.14-alpine; Redis 7.4-alpine; Celery
Authentication fixture	Temporary local RS256/JWKS issuer; normal application signature/claim verification
Scan implementation	Local file storage; asynchronous Celery worker using <code>stub-v0</code> classification
Performance	Grafana k6 2.3.0; isolated loopback HTTP target
Browser UI	Playwright Python 1.63.0; headless Google Chrome 153.0.8010.47; Next.js 16.2.12 dev server
Browser viewports	Desktop 1440 × 900; narrow viewport 390 × 844

2.3 Dependencies, assumptions and exclusions

Tests use disposable local data and synthetic tenant identities. Live stack scripts require Docker and the Python dependencies; browser checks require the web app, configured public Supabase settings and Chrome. The browser checks use no authenticated account. Invalid form submissions are rejected before authentication, and no real password is entered.

3. Test layering and techniques

3.1 Classification and execution

Layer	Technique and tools	Execution represented here
Unit	Controlled inputs and assertions; pytest or node : test/tsx	API business logic, ML utilities, web/mobile utilities
API/component	FastAPI TestClient with substituted database, storage or queue dependencies	Request/response and orchestration checks
Database integration	Real PostgreSQL migrations, RLS and restricted-role SQL	Two security suites with ON_ERROR_STOP=1
Service integration	Live HTTP plus real PostgreSQL, Redis and Celery	14 local cases; local issuer and stub classifier
Browser UI	Playwright interacts with the real rendered Next.js app	Eight local browser smoke cases
Performance	k6 virtual users and explicit pass/fail thresholds	Smoke, load and spike profiles
Static analysis	ESLint and TypeScript compilation checks	Web lint; web and mobile typechecks

The existing GitHub Actions workflow runs repository structure, API, ML, web, mobile and RLS jobs on pull requests and pushes to main. The new live integration, k6 and Playwright evidence was collected locally; this report does not claim that those suites already run in CI or that these artifacts were uploaded by CI.

3.2 Test design and success criteria

Positive cases exercise valid requests and expected state changes. Negative cases exercise missing/malformed identity, unauthorized roles, invalid uploads, cross-tenant access, overselling and conflicting idempotency payloads. Boundary cases exercise confidence cutoffs and depleted stock. Concurrency cases overlap transactions against the real database. Browser cases assert visible state and navigation after user actions.

Functional tests pass when every assertion succeeds and the command exits successfully. SQL execution stops on the first error. Performance profiles must satisfy **all** configured thresholds; passing HTTP-status checks alone is insufficient. Static analysis is supporting quality evidence, not a runtime functional test.

3.3 Coverage interpretation

Requirement traceability is provided by the cases and evidence references below. Test counts represent executed examples, not the percentage of requirements verified. No line/branch coverage instrument was enabled for these runs. Frontend utility tests cannot establish React rendering, and ML unit tests with controlled predictions cannot establish model accuracy.

This submission does not measure trained YOLO accuracy, real inference latency, phone camera capture, offline behaviour, iOS/Android device compatibility, full authenticated dashboard journeys, accessibility conformance, hosted Supabase availability, R2 behaviour or disaster recovery. Storage in the live scan test is local disk. Short laptop runs are not production capacity or endurance benchmarks. No user acceptance study or code-coverage percentage is claimed.

4. Detailed functional test cases

4.1 API-01 — Scan upload and orchestration

Purpose: validate scan inputs and asynchronous processing. **Setup:** FastAPI TestClient with controlled roles and substituted dependencies. **Procedure:** submit valid/invalid images and quantities; exercise retrieval, listing, publication failure and Redis keys. **Expected:** valid creation returns HTTP 202 without inline inference; invalid inputs and forbidden roles are rejected; enqueue failures mark the scan failed; keys carry a tenant namespace. **Result:** 10 passed (E01), included in the API total. Camera and cloud storage are excluded; INT-01 covers the live pipeline.

4.2 API-02 — Sales integrity and idempotency

Purpose: prevent overselling and duplicate deductions. **Setup:** controlled inventory and a fake database. **Procedure:** create valid/excessive sales, mismatch product and batch, omit the key, repeat identical payloads and reuse a key with changed data. **Expected:** valid sales deduct stock and emit applicable alerts; identical replay returns the existing sale without another deduction; changed payloads conflict; forbidden roles are rejected. **Result:** 9 passed (E02), included in the API total. The live contract returns HTTP 201 for original/identical replay and 409 for changed-payload reuse. INT-01 verifies concurrency.

4.3 SEC-01 — Tenant isolation and restricted runtime access

Purpose: prevent access outside the tenant context. **Setup:** disposable PostgreSQL 16, migrations, synthetic tenants and restricted login `freshlens_api_local`. **Procedure:** run `rls_isolation.sql` and `runtime_role.sql`, including restricted login over TCP. **Expected:** tenant boundaries and auth-hook rules hold; runtime privileges exclude superuser and BYPASSRLS while permitting authorized operations. **Result:** both suites passed, exit 0 (E03). Coverage is limited to the exercised policies and operations.

4.4 ML-01 — Identity and freshness confidence gates

Purpose: prevent uncertain classification results from being accepted. **Setup:** controlled fake model outputs in `test_identity.py`. **Procedure:** map supported labels, reject unknown labels, accept identity 0.91 and freshness 0.84, reject identity 0.70 under the 0.75 cutoff, and reject freshness 0.49 under the 0.50 cutoff. **Expected:** low identity confidence leaves identity/freshness labels absent and skips the freshness model; low freshness confidence raises the expected error. **Result:** 5 passed (E04), included in the 28-case ML total. Actual image quality, checkpoints and model accuracy were not evaluated by these cases.

4.5 UI-01 — Web mapping and claims unit tests

Purpose: validate aggregate data transformations and trusted authentication claims. **Procedure:** construct admin API responses and claim objects, then assert percentages, snapshot mapping, role/tenant validation and treatment of user metadata. **Expected:** valid identities map correctly; malformed or incompatible role/tenant combinations are rejected; user metadata is presentation-only. **Result:** 8 passed using `node: test through tsx` (E05). The historical terminal label “UI-01” refers to utility unit tests; it is not evidence that browser charts rendered. Playwright browser coverage is UI-02 in Section 7.

4.6 MOB-01 and supporting regression checks

Mobile tests exercise UTF-8 chunk-size limits, session round trips, atomic replacement/removal of storage chunks and vendor claim validation. **Result:** 9 passed (E06). The complete API and ML suites passed **68 and 28 cases**, respectively (E07–E08). Web lint, web typecheck and mobile typecheck all exited successfully (E09). The API suite reported one Starlette/AnyIO deprecation warning; it did not fail the suite.

5. INT-01 — Live service integration

5.1 Setup and method

tests/system/test_live_api.py exercises the running application over HTTP, with a real PostgreSQL database, Redis and Celery worker. Services bind locally and are discarded after the run. JWT verification uses the normal application implementation with a local RS256/JWKS fixture. The API and worker use the restricted database role. No FastAPI dependency overrides are used in this suite.

The scan test sends a multipart upload, verifies queue/worker completion and persisted inventory, and checks that another tenant cannot retrieve the scan. The classifier is stub-v0; this establishes integration of the services while excluding trained-model behaviour.

5.2 Executed scenarios

Area	Assertions exercised
Authentication and roles	Missing, malformed and tampered token rejection; vendor/admin endpoint permissions
Database boundary	Restricted connection privileges; tenant catalogue isolation over HTTP
Sales and replay	Persistence, identical replay and changed-payload conflict without extra deduction
Atomicity	Cross-tenant write rejection; multi-item oversell rolls back the complete sale
Concurrency	Two sales cannot oversell the last item; concurrent identical requests return one sale for initial stock 1 and 10; conflicting payloads return 409
Key scope and alerts	The same key remains independent between tenants; low-stock alerts persist and can be read
Scan workflow	Upload → local storage → Redis → Celery → database completion; tenant isolation

Result: 14 passed in 2.49 seconds, exit 0 (E10). The total includes two parameterized stock cases for concurrent identical retries.

5.3 Defect discovered and resolved

Concurrent identical sale requests initially reproduced a failure: one request returned HTTP 500 after a database uniqueness conflict left its transaction aborted. The sales service now takes a PostgreSQL transaction advisory lock scoped to tenant and idempotency key before checking for an existing sale and validating stock.

The regression test deliberately overlaps the requests using a real database row lock, then verifies both responses identify the same sale and stock is deducted once. All 14 integration cases and the existing 68 API cases passed after the change. The source fix remains local and uncommitted in this evidence revision.

6. Performance testing with Grafana k6

6.1 Workload and acceptance criteria

Each profile starts a new isolated local stack. Each of two tenants has one product, one batch containing 1,000,000 stock units, 20 completed scan records and five alerts. Every iteration reads products, batches, alerts and scans. Every fifth iteration creates a sale and immediately repeats it with the same idempotency key. Each virtual user pauses 250 ms per iteration.

Checks assert successful statuses, nonempty lists, tenant ownership and identical sale IDs on replay. A post-run database assertion verifies remaining inventory equals starting stock minus persisted sales. k6 uses sequential requests within each virtual user; VU count is not a fixed request-rate target. The one-VU smoke profile exercises tenant A's HTTP workload; tenant B remains a seeded control for its post-run stock check. The load and spike profiles exercise both tenants.

The following **local acceptance criteria were chosen before execution**: read-request p95 below **500 ms**; sale/replay p95 below **1,000 ms**; HTTP failure rate below **1%**; and **100%** functional checks passing. p95 is the 95th percentile of measured request duration. These criteria are test targets, not a contractual production SLA. k6 treats a failed threshold as a failed run [R4].

6.2 Profiles and measured results

ID / profile	Virtual-user schedule		Configured duration
PERF-01 / smoke	1 VU		10 s
PERF-02 / load	10 s to 5 VUs; 10 s to 10; hold 30 s; ramp to 0 in 10 s		60 s
PERF-03 / spike	10 s to 2; 2 s to 25; hold 15 s; 2 s to 2; hold 10 s; 5 s to 0		44 s

Metric	Smoke	Load	Spike
HTTP requests	84	1,754	1,032
Completed iterations	19	397	227
Functional checks passed	240 / 240	5,013 / 5,013	2,910 / 2,910
Read p95 (ms)	77.50	308.80	668.52
Write/replay p95 (ms)	110.91	499.62	1,192.94
HTTP failure rate	0%	0%	0%
Post-run stock checks	Both tenants pass	Both tenants pass	Both tenants pass
Process exit code	0	0	99
Overall profile verdict	PASS	PASS	FAIL — latency

Values are rounded from the retained k6 JSON summaries. Terminal formatting truncates some displayed values, including load write p95 at 499.61 ms. Evidence E11–E19 records setup, thresholds and completion for each run.

6.3 Interpretation and limitations

The spike exceeds the read p95 target by **168.52 ms** and write p95 target by **192.94 ms**. All functional assertions still pass and no HTTP requests fail, but the performance verdict remains **FAIL**. The observed throughput and latency describe this particular short local workload; they do not identify a production user limit or prove the cause of the degradation. Bottleneck profiling, optimization and another measured run remain follow-up work.

The screenshots show an “unknown field gracefulStop” warning. k6 ignored the attempted top-level setting and used its **default 30-second graceful-stop period**, as printed in the scenario output. All iterations completed with zero interruptions. The warning is preserved in the evidence; the report records the effective configuration rather than the ignored setting.

The API, database, load generator and other applications shared the laptop. There was one API worker and a small synthetic dataset. No CPU, memory or database-pool profiling was recorded. The k6 workload does not upload scan images, run real ML inference, measure browser rendering or establish soak/endurance behaviour. The aggregate spike summary also does not independently prove how quickly latency recovered after the peak.

7. UI-02 — Browser testing with Playwright

Playwright was used for UI testing of the real Next.js administration sign-in interface. Python Playwright drove headless Chrome against the local development server; pytest recorded the assertions and JUnit results. Each case used a new browser context. No authenticated user or mocked API response was needed for this smoke scope.

Browser case	Expected behaviour	Result
Login page and controls	Correct title/heading; labelled email/password controls; enabled submit; password masked	PASS
Empty submission	Both field errors visible; aria-invalid=true; remains on login	PASS
Malformed email	Email error appears before authentication; password is not marked invalid	PASS
Password visibility	Show/hide changes input type and preserves the entered synthetic value	PASS
Protected dashboard	Unauthenticated navigation redirects to login	PASS
Session-expired route	Redirects to login with the session-expiry alert	PASS
Narrow layout	At 390 × 844, submit remains reachable and inside viewport width; no horizontal overflow	PASS
Access-denied navigation	Required-role heading renders; return link opens login	PASS

Recorded result: 8 passed in 12.79 seconds. Test source: tests/ui/test_web_ui.py. Evidence E20 shows the desktop validation state after an empty submission; E21 shows the narrow layout. Screenshots are supplementary visual evidence; the result log and JUnit XML record the actual assertions.

The browser was Chrome 153.0.8010.47, controlled by Playwright 1.63.0, at desktop 1440 × 900 and narrow 390 × 844 viewports. The run does not establish authenticated dashboard/chart behaviour, full accessibility compliance, cross-browser compatibility or native Expo device behaviour. The narrow viewport is not an iOS/Android device test. The entered example values are synthetic and contain no real credentials.

```

Playwright / pytest execution, 27 September 2026
Collected: 8 browser cases
Result:    8 passed in 12.79s
JUnit:     docs/test-report-assets/playwright-results.xml
Log:       docs/test-report-assets/playwright-results.log

```

8. Evidence, findings and reproducibility

8.1 Evidence handling

The appendix contains **19 distinct user-captured terminal screenshots**, followed by two genuine Playwright browser screenshots. Repeated attachments have been included once. Original PNG content is preserved; images are only scaled to fit report pages. Each figure has an evidence ID, source filename, scope and result caption. docs/test-report-assets/manifest.json records SHA-256 hashes for all 21 images.

Original terminal results are retained under runs/test-evidence/terminal-20260927-185722/. Integration/k6 evidence is under runs/test-evidence/terminal-20260927-212122/, with per-profile summaries, JUnit where applicable, compressed raw metric samples and command logs/exit codes. Samples are timing/status metrics, not a full HTTP header/body capture. Playwright artifacts are retained under runs/test-evidence/playwright-ui/; its log and JUnit are also copied beside the report assets. Temporary local tokens and database connection configuration are excluded from the submission bundle.

8.2 Findings and remaining work

Finding	Status and required follow-up
Concurrent identical sales could return HTTP 500	Fixed locally with a tenant/key advisory lock; integration and API regression tests pass
Spike read and write p95 exceed local limits	Open performance finding; profile resource/connection contention, optimize and rerun
k6 graceful-stop configuration warning	Recorded run used the default 30 s; correct configuration placement before future runs
Starlette/AnyIO deprecation warning	Non-failing dependency warning; review compatibility during dependency maintenance
Coverage gaps	Authenticated browser flows, native mobile/device behaviour, trained-model evaluation and production-scale testing remain outside this evidence

8.3 Reproduction commands

Run from the repository root after installing the API/ML requirements in the Python 3.12 test environment and installing web/mobile npm dependencies:

```
# Original suites and SQL checks in screenshot-ready terminal tabs
python3 scripts/test-evidence-terminal.py
# Live integration, k6 smoke, load and spike, sequentially
python3 scripts/test-evidence-terminal.py --system
# Browser tests: start the local Next.js app in another terminal
npm --prefix apps/web run dev -- --hostname 127.0.0.1 --port 3107
.venv-test-report/bin/python -m pytest tests/ui/test_web_ui.py -v \
  --junitxml=runs/test-evidence/playwright-ui/results.xml
# Rebuild this report from Markdown and the evidence manifest
.venv-test-report/bin/python scripts/build-test-report.py
```

9. Implemented inventory and references

9.1 Test inventory

Location	Files / scope
apps/ api/ tests/	test_admin.py, test_auth.py, test_authorization.py, test_config.py, test_health.py, test_sales.py, test_scans.py, test_storage.py, test_tenant_context.py
package s/ml/ tests/	test_dataset_snapstock.py, test_evaluate_freshness.py, test_identity.py, test_imagenet_produce.py, test_inventory_batch.py, test_prepare_freshness_dataset.py, test_prepare_identity_dataset.py, test_stub_classifier.py
Web unit tests	src/lib/auth/claims.test.ts, src/lib/api/admin-map.test.ts under apps/web/
Mobile unit tests	src/lib/auth/claims.test.ts, src/lib/auth/chunked-storage.test.ts under apps/mobile/
infra/ db/ tests/	rls_isolation.sql, runtime_role.sql
tests/ system/	test_live_api.py, conftest.py, local_stack.py
tests/ perform ance/	api.k6.js — all three performance profiles
tests/ ui/	test_web_ui.py — eight Playwright browser cases

9.2 References

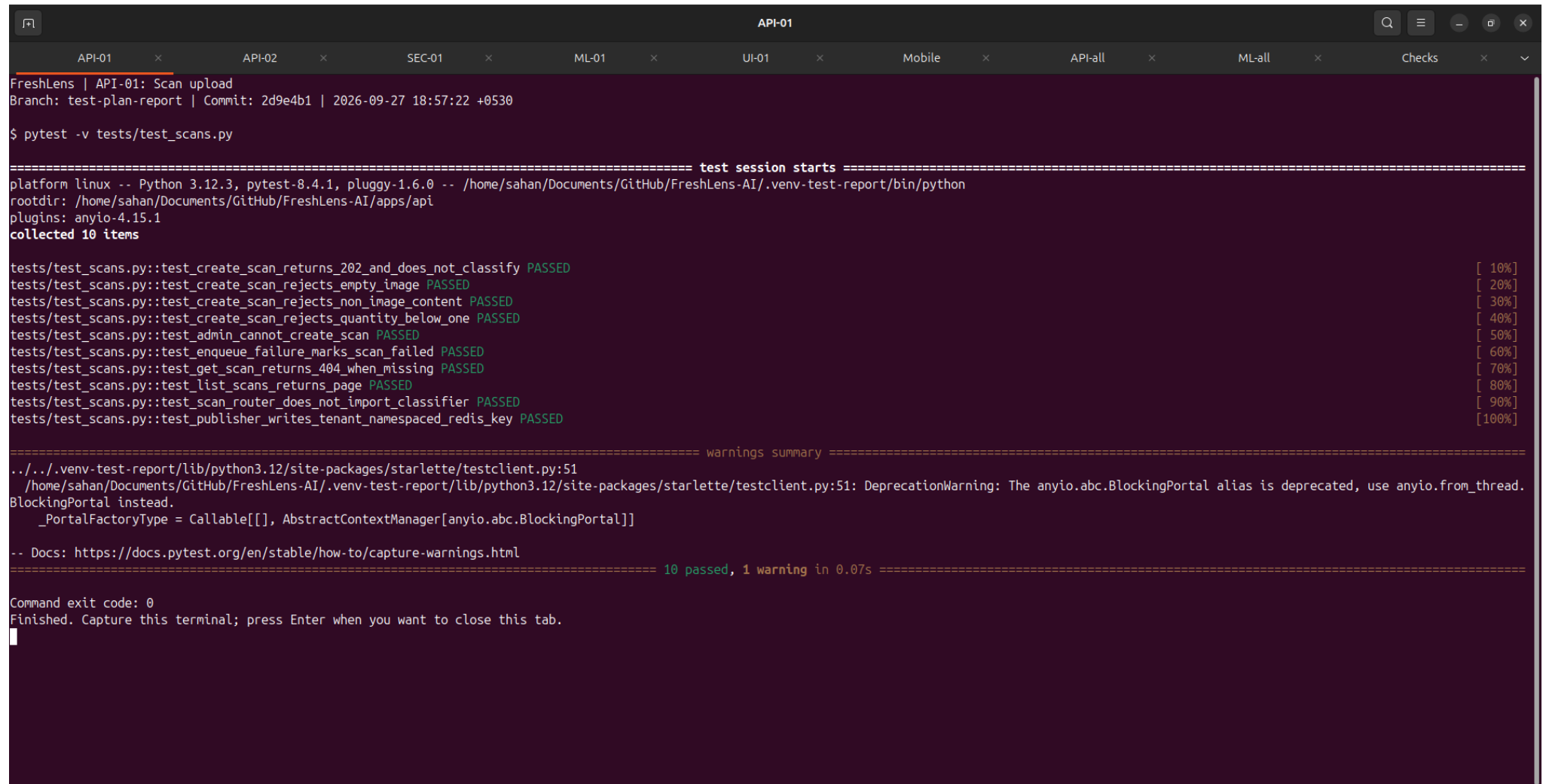
- **R1.** FreshLens API contract: docs/api/v1/openapi.yaml.
- **R2.** FreshLens architecture and authentication: docs/design/FreshLens-SAD.md and docs/authentication.md.
- **R3.** Local execution records: original terminal run terminal-20260927-185722; integration/k6 run terminal-20260927-212122; Playwright results.xml and results.log. These records support the measured outcomes in this report.
- **R4.** Grafana, k6 thresholds. Defines threshold-based performance verdicts; accessed 27 September 2026.
- **R5.** Microsoft, Playwright for Python. Browser automation and Python testing documentation; accessed 27 September 2026.

Appendix A. Screenshot evidence

Figures E01–E19 are the supplied terminal captures in test order. Figures E20–E21 are the additional Playwright browser captures. The appendix uses full-size landscape pages for terminal readability and a portrait page for the narrow browser capture. Figure captions describe the scope and do not change the recorded results.

Figure	Test / evidence
E01	API-01: scan request and orchestration tests
E02	API-02: sales business rules
E03	SEC-01: PostgreSQL isolation and restricted role
E04	ML-01: identity and freshness confidence gates
E05	UI-01: web utility unit tests
E06	MOB-01: mobile authentication and session storage
E07	Complete API regression suite
E08	Complete ML utility and worker-logic suite
E09	Static analysis: lint and TypeScript
E10	INT-01: live service integration
E11	PERF-01: k6 smoke configuration
E12	PERF-01: smoke thresholds and checks
E13	PERF-01: smoke metrics and completion
E14	PERF-02: k6 load configuration
E15	PERF-02: load thresholds and checks
E16	PERF-02: load metrics and completion
E17	PERF-03: k6 spike configuration
E18	PERF-03: spike latency threshold failures
E19	PERF-03: spike metrics and failed completion
E20	UI-02: Playwright desktop form validation
E21	UI-02: Playwright mobile-width layout

E01 — API-01: scan request and orchestration tests



The image shows a terminal window titled 'API-01' with a tab bar at the top containing 'API-01', 'API-02', 'SEC-01', 'ML-01', 'UI-01', 'Mobile', 'API-all', 'ML-all', and 'Checks'. The terminal output shows the command `$ pytest -v tests/test_scans.py` and the following results:

```
===== test session starts =====
platform linux -- Python 3.12.3, pytest-8.4.1, pluggy-1.6.0 -- /home/sahan/Documents/GitHub/FreshLens-AI/.venv-test-report/bin/python
rootdir: /home/sahan/Documents/GitHub/FreshLens-AI/apps/api
plugins: anyio-4.15.1
collected 10 items

tests/test_scans.py::test_create_scan_returns_202_and_does_not_classify PASSED [ 10%]
tests/test_scans.py::test_create_scan_rejects_empty_image PASSED [ 20%]
tests/test_scans.py::test_create_scan_rejects_non_image_content PASSED [ 30%]
tests/test_scans.py::test_create_scan_rejects_quantity_below_one PASSED [ 40%]
tests/test_scans.py::test_admin_cannot_create_scan PASSED [ 50%]
tests/test_scans.py::test_enqueue_failure_marks_scan_failed PASSED [ 60%]
tests/test_scans.py::test_get_scan_returns_404_when_missing PASSED [ 70%]
tests/test_scans.py::test_list_scans_returns_page PASSED [ 80%]
tests/test_scans.py::test_scan_router_does_not_import_classifier PASSED [ 90%]
tests/test_scans.py::test_publisher_writes_tenant_namespaced_redis_key PASSED [100%]

===== warnings summary =====
.../.venv-test-report/lib/python3.12/site-packages/starlette/testclient.py:51
/home/sahan/Documents/GitHub/FreshLens-AI/.venv-test-report/lib/python3.12/site-packages/starlette/testclient.py:51: DeprecationWarning: The anyio.abc.BlockingPortal alias is deprecated, use anyio.from_thread.
BlockingPortal instead.
  _PortalFactoryType = Callable[[], AbstractContextManager[anyio.abc.BlockingPortal]]

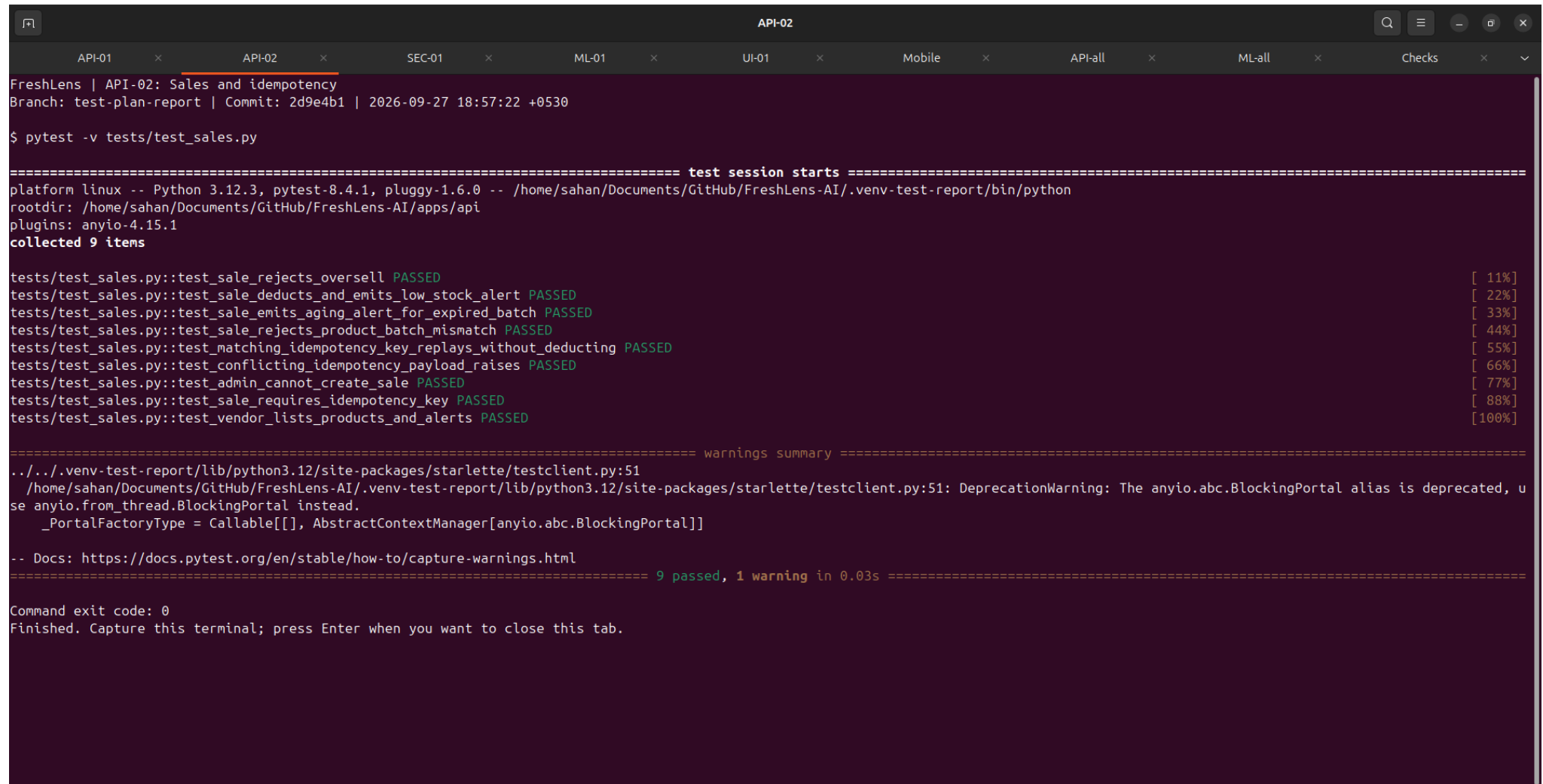
-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
===== 10 passed, 1 warning in 0.07s =====

Command exit code: 0
Finished. Capture this terminal; press Enter when you want to close this tab.
```

E01. 10 passed; multipart input validation, asynchronous acceptance and tenant-namespaced publishing. Component tests use substituted dependencies.

Source: Screenshot from 2026-09-27 18-58-00.png · User-captured terminal screenshot

E02 — API-02: sales business rules



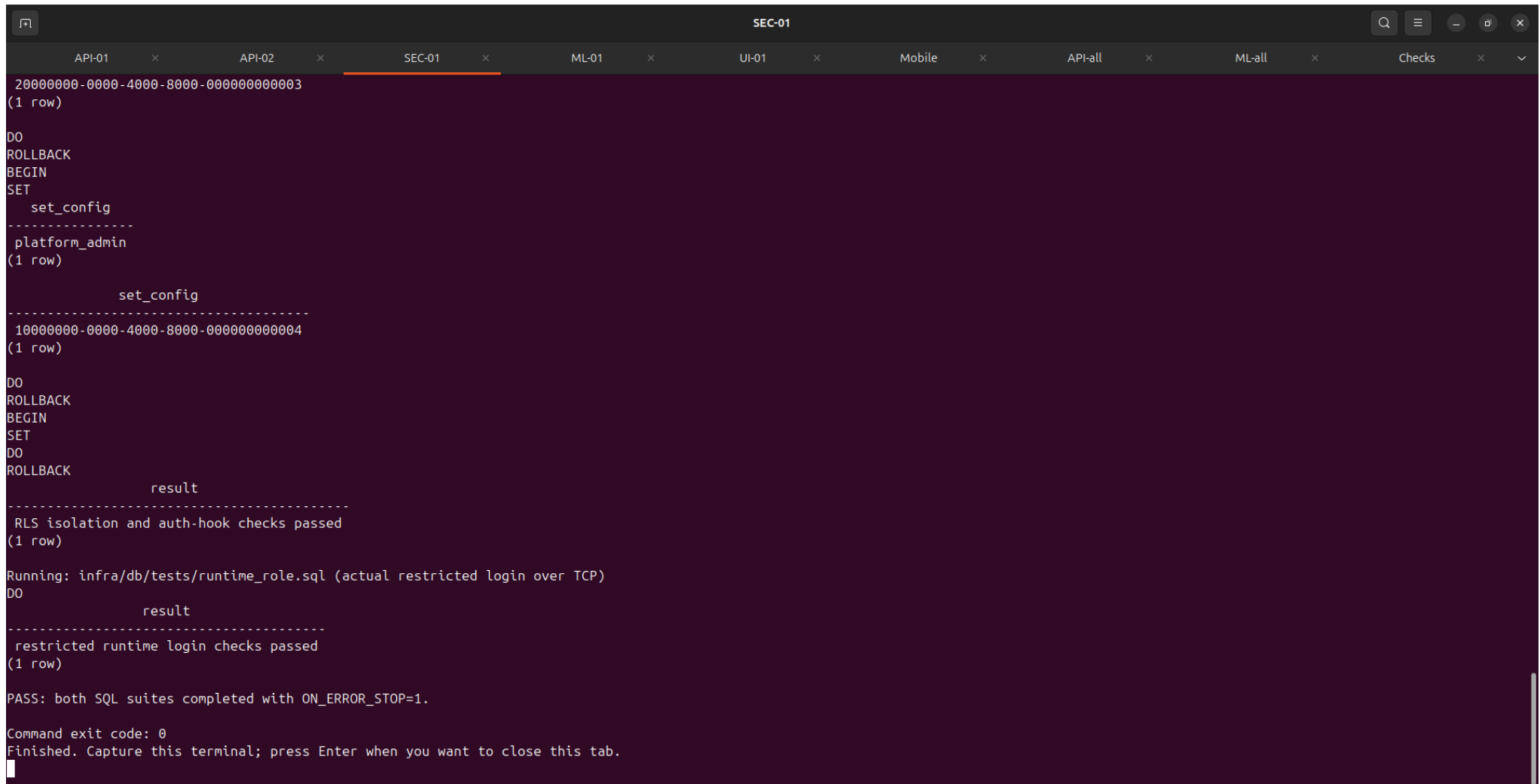
The image is a terminal window titled "API-02" showing the execution of pytest tests. The terminal output includes the following information:

- Branch: test-plan-report | Commit: 2d9e4b1 | 2026-09-27 18:57:22 +0530
- Command: `$ pytest -v tests/test_sales.py`
- Test session starts with platform linux, Python 3.12.3, pytest-8.4.1, and pluggy-1.6.0.
- 9 items collected.
- Test results (all passed):
 - `tests/test_sales.py::test_sale_rejects_oversell` PASSED [11%]
 - `tests/test_sales.py::test_sale_deducts_and_emits_low_stock_alert` PASSED [22%]
 - `tests/test_sales.py::test_sale_emits_aging_alert_for_expired_batch` PASSED [33%]
 - `tests/test_sales.py::test_sale_rejects_product_batch_mismatch` PASSED [44%]
 - `tests/test_sales.py::test_matching_idempotency_key_replays_without_deducting` PASSED [55%]
 - `tests/test_sales.py::test_conflicting_idempotency_payload_raises` PASSED [66%]
 - `tests/test_sales.py::test_admin_cannot_create_sale` PASSED [77%]
 - `tests/test_sales.py::test_sale_requires_idempotency_key` PASSED [88%]
 - `tests/test_sales.py::test_vendor_lists_products_and_alerts` PASSED [100%]
- Warnings summary: A DeprecationWarning is shown about the `anyio.abc.BlockingPortal` alias being deprecated.
- Final summary: `9 passed, 1 warning in 0.03s`
- Command exit code: 0

E02. 9 passed; stock rules and sequential idempotency checks with a fake database. Live concurrency is covered separately by INT-01.

Source: Screenshot from 2026-09-27 18-58-05.png · User-captured terminal screenshot

E03 — SEC-01: PostgreSQL isolation and restricted role

A terminal window titled 'SEC-01' with a dark background and light green text. The window has a tab bar at the top with tabs for 'API-01', 'API-02', 'SEC-01' (active), 'ML-01', 'UI-01', 'Mobile', 'API-all', 'ML-all', and 'Checks'. The terminal output shows two SQL test suites. The first suite, 'platform_admin', has a result of '20000000-0000-4000-8000-000000000003 (1 row)'. The second suite, 'set_config', has a result of '10000000-0000-4000-8000-000000000004 (1 row)'. Both suites are marked as 'PASS: both SQL suites completed with ON_ERROR_STOP=1.' and the command exit code is 0. The terminal also shows 'Running: infra/db/tests/runtime_role.sql (actual restricted login over TCP)' and 'restricted runtime login checks passed (1 row)'.

```
SEC-01
20000000-0000-4000-8000-000000000003
(1 row)

DO
ROLLBACK
BEGIN
SET
  set_config
-----
platform_admin
(1 row)

      set_config
-----
10000000-0000-4000-8000-000000000004
(1 row)

DO
ROLLBACK
BEGIN
SET
DO
ROLLBACK
      result
-----
RLS isolation and auth-hook checks passed
(1 row)

Running: infra/db/tests/runtime_role.sql (actual restricted login over TCP)
DO
      result
-----
restricted runtime login checks passed
(1 row)

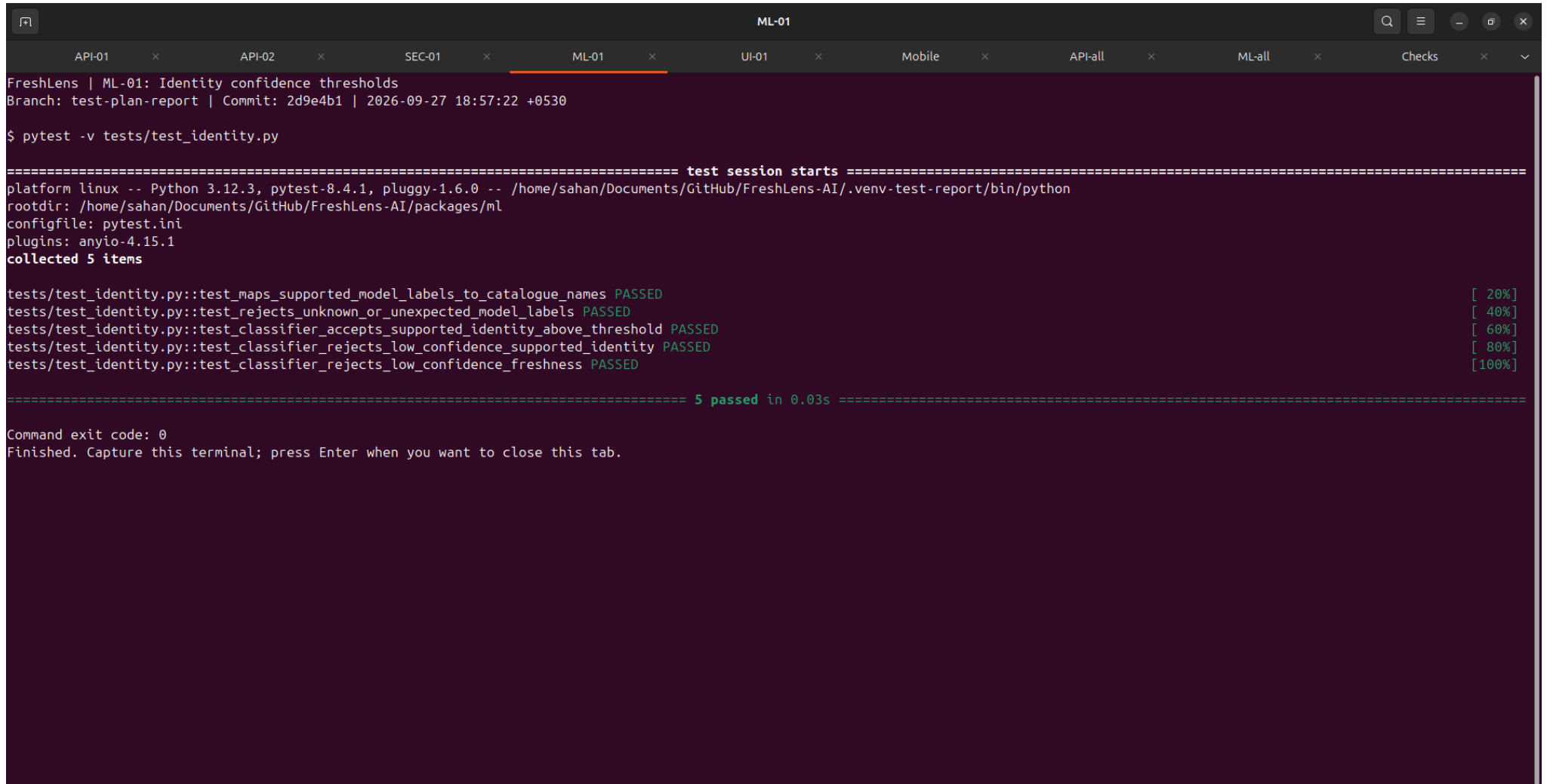
PASS: both SQL suites completed with ON_ERROR_STOP=1.

Command exit code: 0
Finished. Capture this terminal; press Enter when you want to close this tab.
```

E03. Both SQL suites passed with ON_ERROR_STOP=1; the final command exit code is 0.

Source: Screenshot from 2026-09-27 18-58-10.png · User-captured terminal screenshot

E04 — ML-01: identity and freshness confidence gates



The image is a terminal window titled "ML-01" with a dark theme. It shows the execution of a pytest command and its output. The terminal has a tab bar at the top with tabs for API-01, API-02, SEC-01, ML-01 (active), UI-01, Mobile, API-all, ML-all, and Checks. The output of the command is as follows:

```
FreshLens | ML-01: Identity confidence thresholds
Branch: test-plan-report | Commit: 2d9e4b1 | 2026-09-27 18:57:22 +0530

$ pytest -v tests/test_identity.py

===== test session starts =====
platform linux -- Python 3.12.3, pytest-8.4.1, pluggy-1.6.0 -- /home/sahan/Documents/GitHub/FreshLens-AI/.venv-test-report/bin/python
rootdir: /home/sahan/Documents/GitHub/FreshLens-AI/packages/ml
configfile: pytest.ini
plugins: anyio-4.15.1
collected 5 items

tests/test_identity.py::test_maps_supported_model_labels_to_catalogue_names PASSED [ 20%]
tests/test_identity.py::test_rejects_unknown_or_unexpected_model_labels PASSED [ 40%]
tests/test_identity.py::test_classifier_accepts_supported_identity_above_threshold PASSED [ 60%]
tests/test_identity.py::test_classifier_rejects_low_confidence_supported_identity PASSED [ 80%]
tests/test_identity.py::test_classifier_rejects_low_confidence_freshness PASSED [100%]

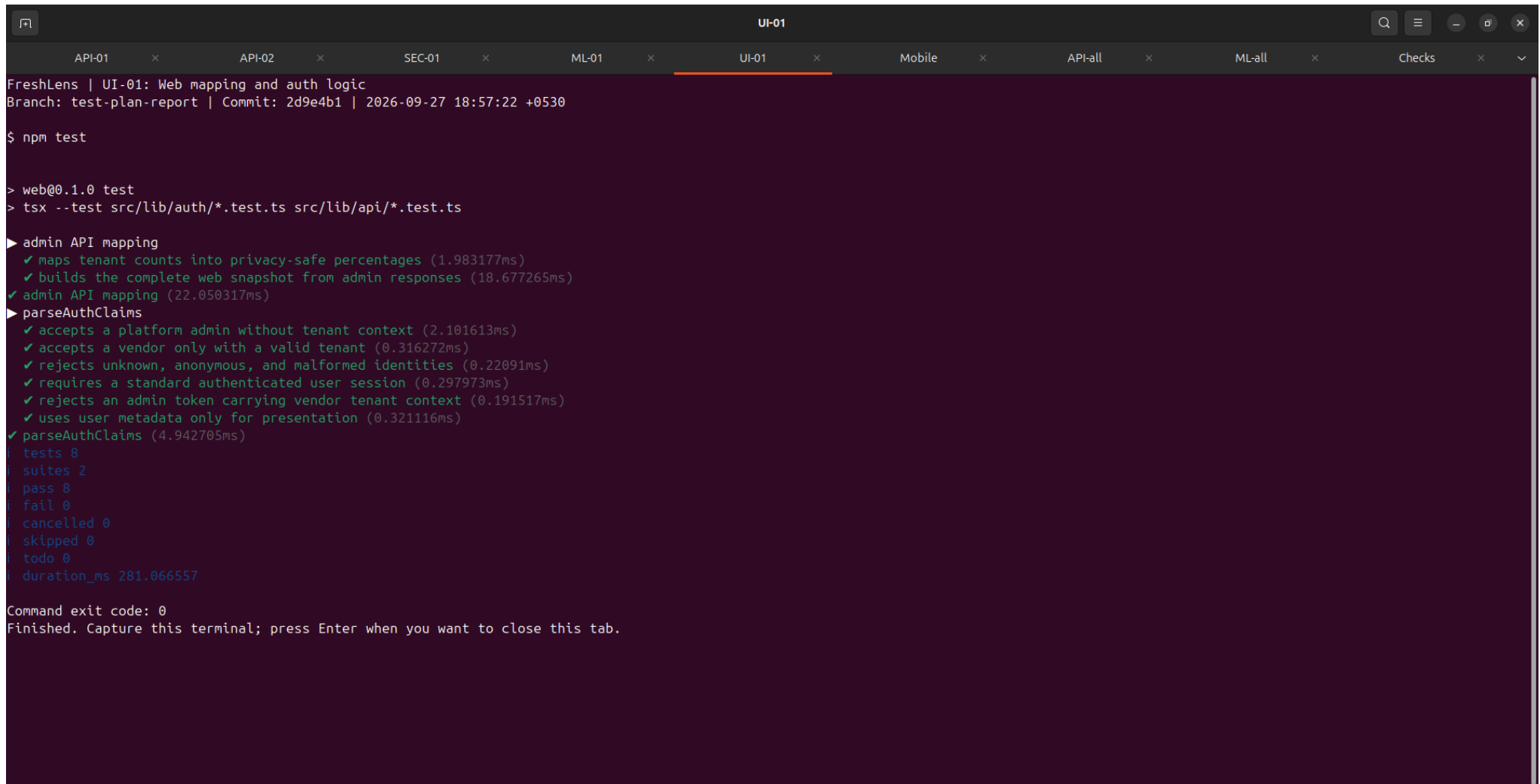
===== 5 passed in 0.03s =====

Command exit code: 0
Finished. Capture this terminal; press Enter when you want to close this tab.
```

E04. 5 passed using controlled model predictions. This verifies decision logic, not trained-model accuracy.

Source: Screenshot from 2026-09-27 18-58-13.png · User-captured terminal screenshot

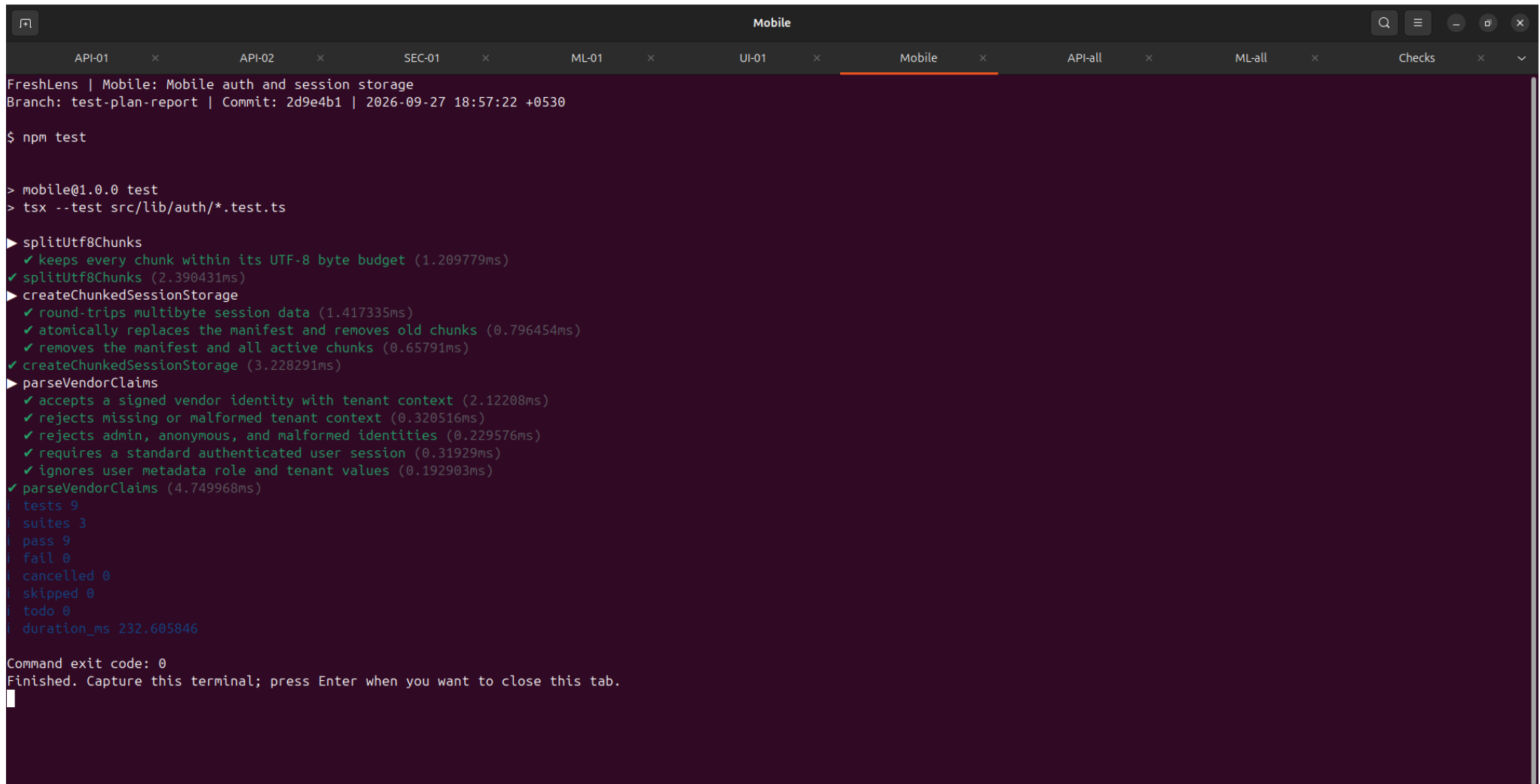
E05 — UI-01: web utility unit tests

A terminal window titled "UI-01" with a dark background and light green text. The window has a tab bar at the top with tabs for API-01, API-02, SEC-01, ML-01, UI-01 (active), Mobile, API-all, ML-all, and Checks. The terminal content shows the output of running 'npm test'. It starts with 'FreshLens | UI-01: Web mapping and auth logic' and 'Branch: test-plan-report | Commit: 2d9e4b1 | 2026-09-27 18:57:22 +0530'. The command '\$ npm test' is executed, followed by 'web@0.1.0 test' and 'tsx --test src/lib/auth/*.test.ts src/lib/api/*.test.ts'. The test results show two main sections: 'admin API mapping' and 'parseAuthClaims'. The 'admin API mapping' section has two passing tests: 'maps tenant counts into privacy-safe percentages (1.983177ms)' and 'builds the complete web snapshot from admin responses (18.677265ms)'. The 'parseAuthClaims' section has six passing tests: 'accepts a platform admin without tenant context (2.101613ms)', 'accepts a vendor only with a valid tenant (0.316272ms)', 'rejects unknown, anonymous, and malformed identities (0.22091ms)', 'requires a standard authenticated user session (0.297973ms)', 'rejects an admin token carrying vendor tenant context (0.191517ms)', and 'uses user metadata only for presentation (0.321116ms)'. The final summary shows 'tests 8', 'suites 2', 'pass 8', 'fail 0', 'cancelled 0', 'skipped 0', 'todo 0', and 'duration_ms 281.066557'. The command exit code is 0, and a message says 'Finished. Capture this terminal; press Enter when you want to close this tab.'

E05. 8 passed using node:test through tsx. These tests cover mappings and authentication claims; browser UI testing is recorded as UI-02.

Source: Screenshot from 2026-09-27 18-58-20.png · User-captured terminal screenshot

E06 — MOB-01: mobile authentication and session storage



```
Mobile
FreshLens | Mobile: Mobile auth and session storage
Branch: test-plan-report | Commit: 2d9e4b1 | 2026-09-27 18:57:22 +0530

$ npm test

> mobile@1.0.0 test
> tsx --test src/lib/auth/*.test.ts

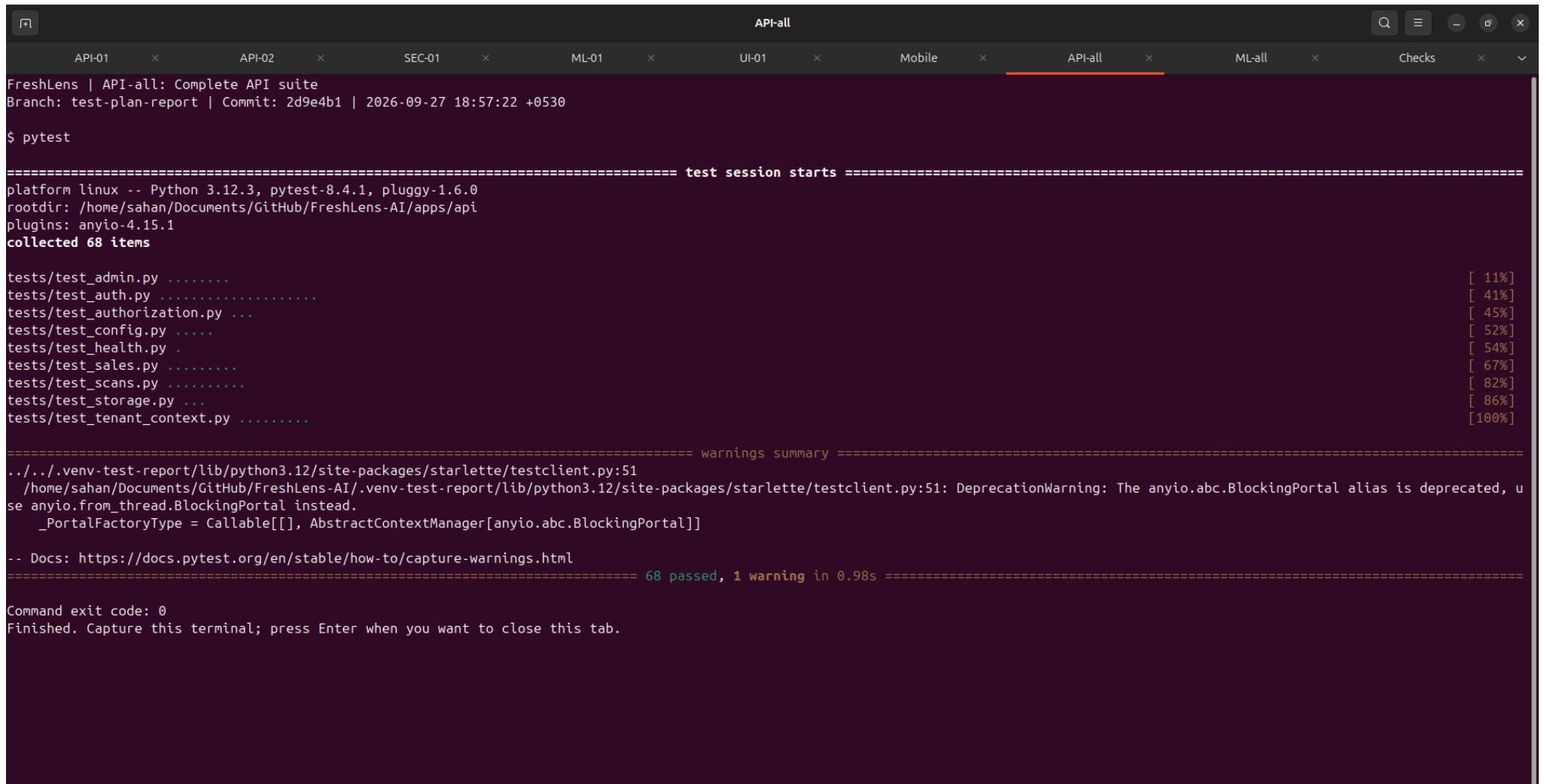
▶ splitUtf8Chunks
  ✓ keeps every chunk within its UTF-8 byte budget (1.209779ms)
✓ splitUtf8Chunks (2.390431ms)
▶ createChunkedSessionStorage
  ✓ round-trips multibyte session data (1.417335ms)
  ✓ atomically replaces the manifest and removes old chunks (0.796454ms)
  ✓ removes the manifest and all active chunks (0.65791ms)
✓ createChunkedSessionStorage (3.228291ms)
▶ parseVendorClaims
  ✓ accepts a signed vendor identity with tenant context (2.12208ms)
  ✓ rejects missing or malformed tenant context (0.320516ms)
  ✓ rejects admin, anonymous, and malformed identities (0.229576ms)
  ✓ requires a standard authenticated user session (0.31929ms)
  ✓ ignores user metadata role and tenant values (0.192903ms)
✓ parseVendorClaims (4.749968ms)
| tests 9
| suites 3
| pass 9
| fail 0
| cancelled 0
| skipped 0
| todo 0
| duration_ms 232.605846

Command exit code: 0
Finished. Capture this terminal; press Enter when you want to close this tab.
```

E06. 9 passed; signed-claim validation and chunked storage logic. This is not phone-device UI automation.

Source: Screenshot from 2026-09-27 18-58-27.png · User-captured terminal screenshot

E07 — Complete API regression suite



```
API-all
FreshLens | API-all: Complete API suite
Branch: test-plan-report | Commit: 2d9e4b1 | 2026-09-27 18:57:22 +0530

$ pytest

===== test session starts =====
platform linux -- Python 3.12.3, pytest-8.4.1, pluggy-1.6.0
rootdir: /home/sahan/Documents/GitHub/FreshLens-AI/apps/api
plugins: anyio-4.15.1
collected 68 items

tests/test_admin.py ..... [ 11%]
tests/test_auth.py ..... [ 41%]
tests/test_authorization.py ... [ 45%]
tests/test_config.py ..... [ 52%]
tests/test_health.py . [ 54%]
tests/test_sales.py ..... [ 67%]
tests/test_scans.py ..... [ 82%]
tests/test_storage.py ... [ 86%]
tests/test_tenant_context.py ..... [100%]

===== warnings summary =====
.../.venv-test-report/lib/python3.12/site-packages/starlette/testclient.py:51
/home/sahan/Documents/GitHub/FreshLens-AI/.venv-test-report/lib/python3.12/site-packages/starlette/testclient.py:51: DeprecationWarning: The anyio.abc.BlockingPortal alias is deprecated, use anyio.from_thread.BlockingPortal instead.
  _PortalFactoryType = Callable[[], AbstractContextManager[anyio.abc.BlockingPortal]]

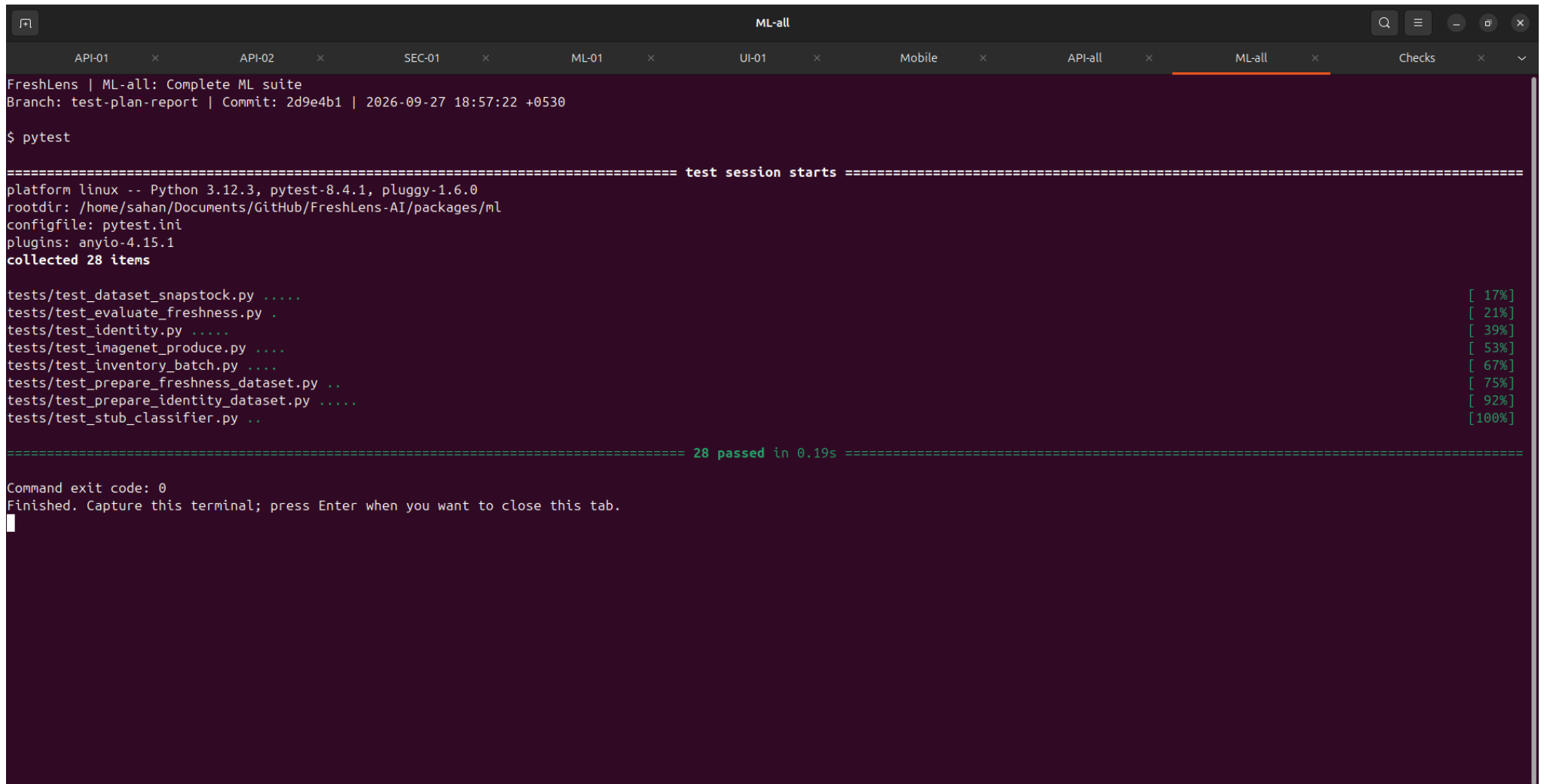
-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
===== 68 passed, 1 warning in 0.98s =====

Command exit code: 0
Finished. Capture this terminal; press Enter when you want to close this tab.
```

E07. 68 passed, with one third-party Starlette/AnyIO deprecation warning; exit code 0. API-01 and API-02 are subsets of this total.

Source: Screenshot from 2026-09-27 18-58-30.png · User-captured terminal screenshot

E08 — Complete ML utility and worker-logic suite



The image shows a terminal window titled "ML-all" with a dark theme. The terminal displays the output of a pytest command. At the top, it shows the branch "test-plan-report" and commit "2d9e4b1". The command "\$ pytest" is entered. The output shows the test session starting, the platform (linux), Python version (3.12.3), and the number of items collected (28). A progress bar shows the progress of the tests, with the following items listed: tests/test_dataset_snapstock.py (17%), tests/test_evaluate_freshness.py (21%), tests/test_identity.py (39%), tests/test_imagenet_produce.py (53%), tests/test_inventory_batch.py (67%), tests/test_prepare_freshness_dataset.py (75%), tests/test_prepare_identity_dataset.py (92%), and tests/test_stub_classifier.py (100%). The final result is "28 passed in 0.19s". The terminal also shows the command exit code (0) and a message to capture the terminal.

```
FreshLens | ML-all: Complete ML suite
Branch: test-plan-report | Commit: 2d9e4b1 | 2026-09-27 18:57:22 +0530

$ pytest

===== test session starts =====
platform linux -- Python 3.12.3, pytest-8.4.1, pluggy-1.6.0
rootdir: /home/sahan/Documents/GitHub/FreshLens-AI/packages/ml
configfile: pytest.ini
plugins: anyio-4.15.1
collected 28 items

tests/test_dataset_snapstock.py ..... [ 17%]
tests/test_evaluate_freshness.py . [ 21%]
tests/test_identity.py ..... [ 39%]
tests/test_imagenet_produce.py .... [ 53%]
tests/test_inventory_batch.py .... [ 67%]
tests/test_prepare_freshness_dataset.py .. [ 75%]
tests/test_prepare_identity_dataset.py ..... [ 92%]
tests/test_stub_classifier.py .. [100%]

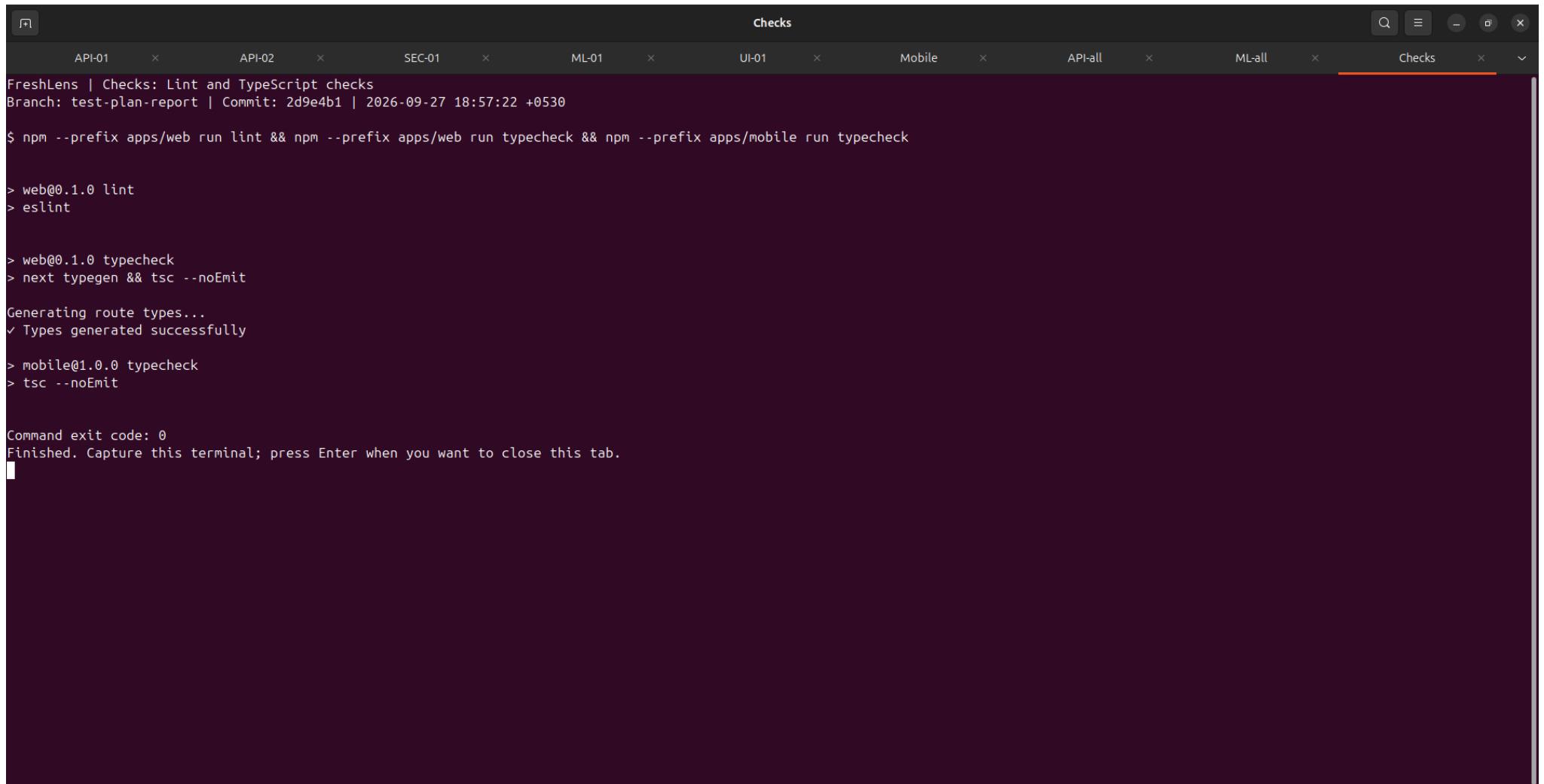
===== 28 passed in 0.19s =====

Command exit code: 0
Finished. Capture this terminal; press Enter when you want to close this tab.
█
```

E08. 28 passed; dataset preparation, confidence logic and inventory/stub utilities. ML-01 is a subset of this total.

Source: Screenshot from 2026-09-27 18-58-33.png · User-captured terminal screenshot

E09 — Static analysis: lint and TypeScript



The image is a terminal window titled "Checks" with a dark background. It shows the execution of lint and TypeScript checks for a web application and a mobile application. The terminal output includes the command to run these checks, the execution of ESLint for the web app, the generation of route types, and the successful completion of TypeScript checks for both the web and mobile apps. The terminal ends with a message indicating the command exited with code 0 and a prompt to capture the terminal.

```
FreshLens | Checks: Lint and TypeScript checks
Branch: test-plan-report | Commit: 2d9e4b1 | 2026-09-27 18:57:22 +0530

$ npm --prefix apps/web run lint && npm --prefix apps/web run typecheck && npm --prefix apps/mobile run typecheck

> web@0.1.0 lint
> eslint

> web@0.1.0 typecheck
> next typegen && tsc --noEmit

Generating route types...
✓ Types generated successfully

> mobile@1.0.0 typecheck
> tsc --noEmit

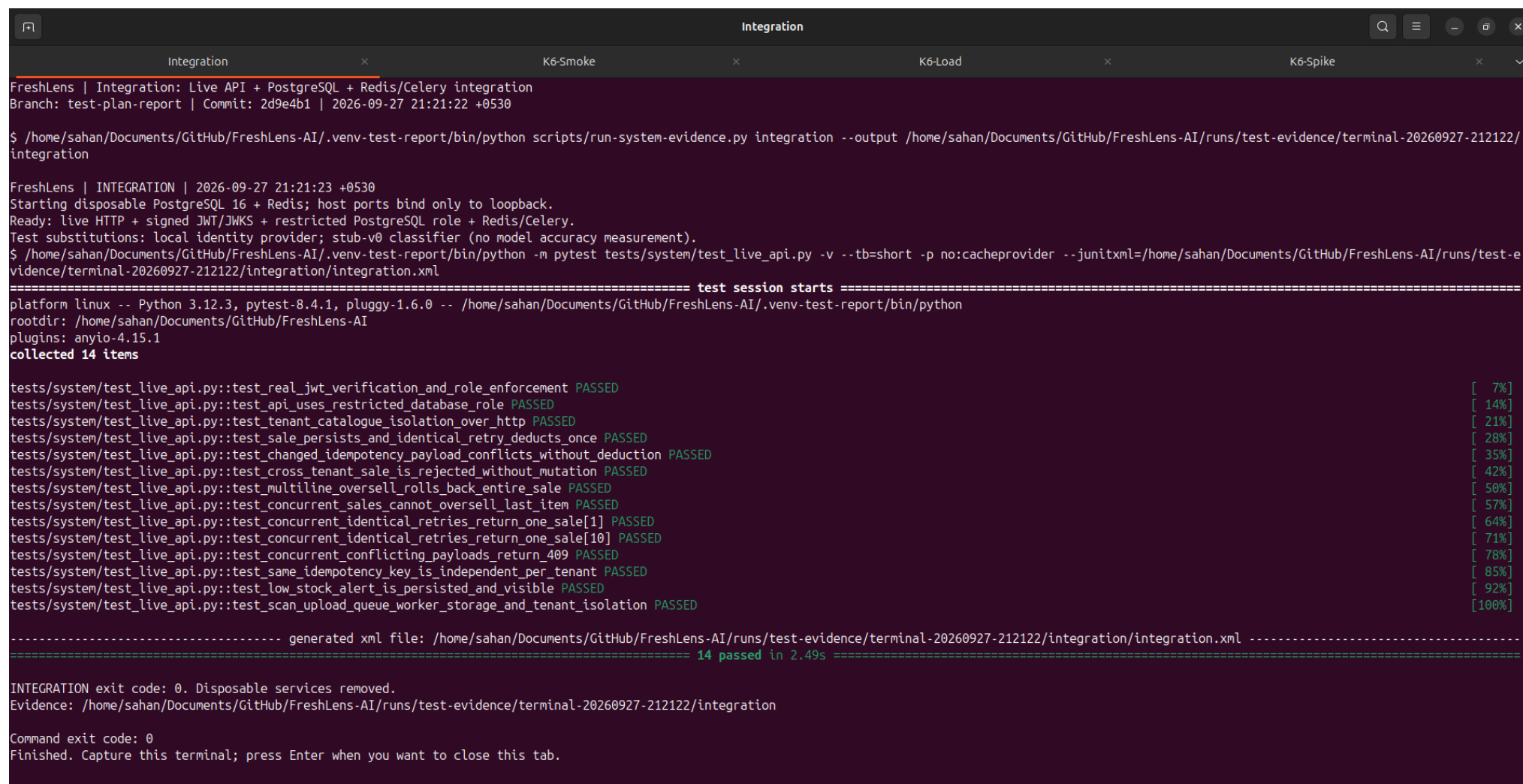
Command exit code: 0
Finished. Capture this terminal; press Enter when you want to close this tab.

```

E09. Web ESLint, web typecheck and mobile typecheck completed successfully; exit code 0. These are static checks rather than runtime tests.

Source: Screenshot from 2026-09-27 18-58-38.png · User-captured terminal screenshot

E10 — INT-01: live service integration



```
FreshLens | Integration: Live API + PostgreSQL + Redis/Celery integration
Branch: test-plan-report | Commit: 2d9e4b1 | 2026-09-27 21:21:22 +0530

$ /home/sahan/Documents/GitHub/FreshLens-AI/.venv-test-report/bin/python scripts/run-system-evidence.py integration --output /home/sahan/Documents/GitHub/FreshLens-AI/runs/test-evidence/terminal-20260927-212122/integration

FreshLens | INTEGRATION | 2026-09-27 21:21:23 +0530
Starting disposable PostgreSQL 16 + Redis; host ports bind only to loopback.
Ready: live HTTP + signed JWT/JWKS + restricted PostgreSQL role + Redis/Celery.
Test substitutions: local identity provider; stub-v0 classifier (no model accuracy measurement).
$ /home/sahan/Documents/GitHub/FreshLens-AI/.venv-test-report/bin/python -m pytest tests/system/test_live_api.py -v --tb=short -p no:cacheprovider --junitxml=/home/sahan/Documents/GitHub/FreshLens-AI/runs/test-evidence/terminal-20260927-212122/integration/integration.xml

===== test session starts =====
platform linux -- Python 3.12.3, pytest-8.4.1, pluggy-1.6.0 -- /home/sahan/Documents/GitHub/FreshLens-AI/.venv-test-report/bin/python
rootdir: /home/sahan/Documents/GitHub/FreshLens-AI
plugins: anyio-4.15.1
collected 14 items

tests/system/test_live_api.py::test_real_jwt_verification_and_role_enforcement PASSED [ 7%]
tests/system/test_live_api.py::test_api_uses_restricted_database_role PASSED [ 14%]
tests/system/test_live_api.py::test_tenant_catalogue_isolation_over_http PASSED [ 21%]
tests/system/test_live_api.py::test_sale_persists_and_identical_retry_deducts_once PASSED [ 28%]
tests/system/test_live_api.py::test_changed_idempotency_payload_conflicts_without_deduction PASSED [ 35%]
tests/system/test_live_api.py::test_cross_tenant_sale_is_rejected_without_mutation PASSED [ 42%]
tests/system/test_live_api.py::test_multiline_oversell_rolls_back_entire_sale PASSED [ 50%]
tests/system/test_live_api.py::test_concurrent_sales_cannot_oversell_last_item PASSED [ 57%]
tests/system/test_live_api.py::test_concurrent_identical_retries_return_one_sale[1] PASSED [ 64%]
tests/system/test_live_api.py::test_concurrent_identical_retries_return_one_sale[10] PASSED [ 71%]
tests/system/test_live_api.py::test_concurrent_conflicting_payloads_return_409 PASSED [ 78%]
tests/system/test_live_api.py::test_same_idempotency_key_is_independent_per_tenant PASSED [ 85%]
tests/system/test_live_api.py::test_low_stock_alert_is_persisted_and_visible PASSED [ 92%]
tests/system/test_live_api.py::test_scan_upload_queue_worker_storage_and_tenant_isolation PASSED [100%]

----- generated xml file: /home/sahan/Documents/GitHub/FreshLens-AI/runs/test-evidence/terminal-20260927-212122/integration/integration.xml -----
===== 14 passed in 2.49s =====

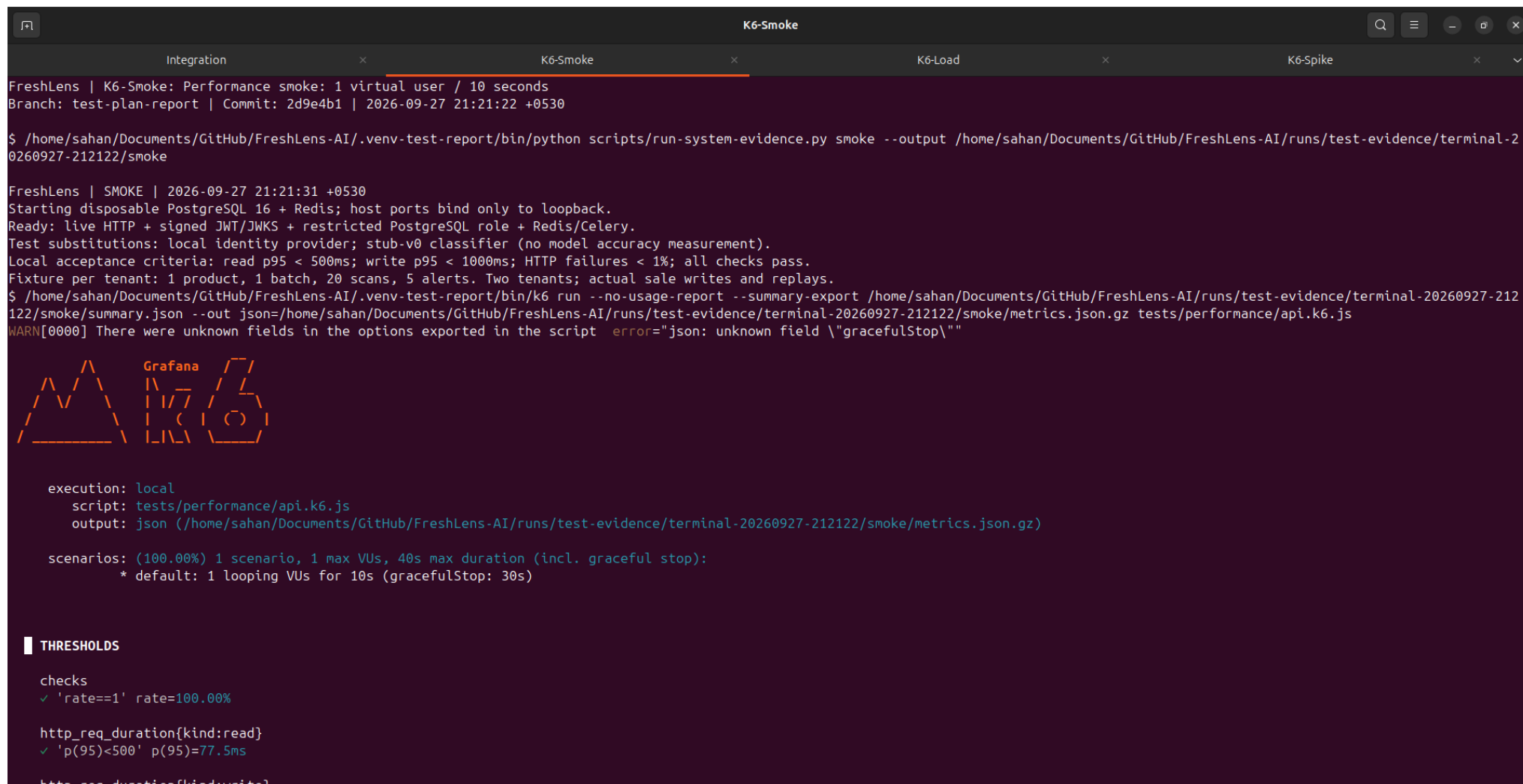
INTEGRATION exit code: 0. Disposable services removed.
Evidence: /home/sahan/Documents/GitHub/FreshLens-AI/runs/test-evidence/terminal-20260927-212122/integration

Command exit code: 0
Finished. Capture this terminal; press Enter when you want to close this tab.
```

E10. 14 passed in 2.49 seconds. Real HTTP, PostgreSQL, Redis and Celery; local JWKS issuer and stub-v0 classifier. Includes the sales concurrency fix.

Source: Screenshot from 2026-09-27 21-35-40.png · User-captured terminal screenshot

E11 — PERF-01: k6 smoke configuration



```
FreshLens | K6-Smoke: Performance smoke: 1 virtual user / 10 seconds
Branch: test-plan-report | Commit: 2d9e4b1 | 2026-09-27 21:21:22 +0530

$ /home/sahan/Documents/GitHub/FreshLens-AI/.venv-test-report/bin/python scripts/run-system-evidence.py smoke --output /home/sahan/Documents/GitHub/FreshLens-AI/runs/test-evidence/terminal-20260927-212122/smoke

FreshLens | SMOKE | 2026-09-27 21:21:31 +0530
Starting disposable PostgreSQL 16 + Redis; host ports bind only to loopback.
Ready: live HTTP + signed JWT/JWKS + restricted PostgreSQL role + Redis/Celery.
Test substitutions: local identity provider; stub-v0 classifier (no model accuracy measurement).
Local acceptance criteria: read p95 < 500ms; write p95 < 1000ms; HTTP failures < 1%; all checks pass.
Fixture per tenant: 1 product, 1 batch, 20 scans, 5 alerts. Two tenants; actual sale writes and replays.
$ /home/sahan/Documents/GitHub/FreshLens-AI/.venv-test-report/bin/k6 run --no-usage-report --summary-export /home/sahan/Documents/GitHub/FreshLens-AI/runs/test-evidence/terminal-20260927-212122/smoke/summary.json --out json=/home/sahan/Documents/GitHub/FreshLens-AI/runs/test-evidence/terminal-20260927-212122/smoke/metrics.json.gz tests/performance/api.k6.js
WARN[0000] There were unknown fields in the options exported in the script  error="json: unknown field \"gracefulStop\""



execution: local
  script: tests/performance/api.k6.js
  output: json (/home/sahan/Documents/GitHub/FreshLens-AI/runs/test-evidence/terminal-20260927-212122/smoke/metrics.json.gz)

scenarios: (100.00%) 1 scenario, 1 max VUs, 40s max duration (incl. graceful stop):
  * default: 1 looping VUs for 10s (gracefulStop: 30s)

THRESHOLDS

checks
✓ 'rate==1' rate=100.00%

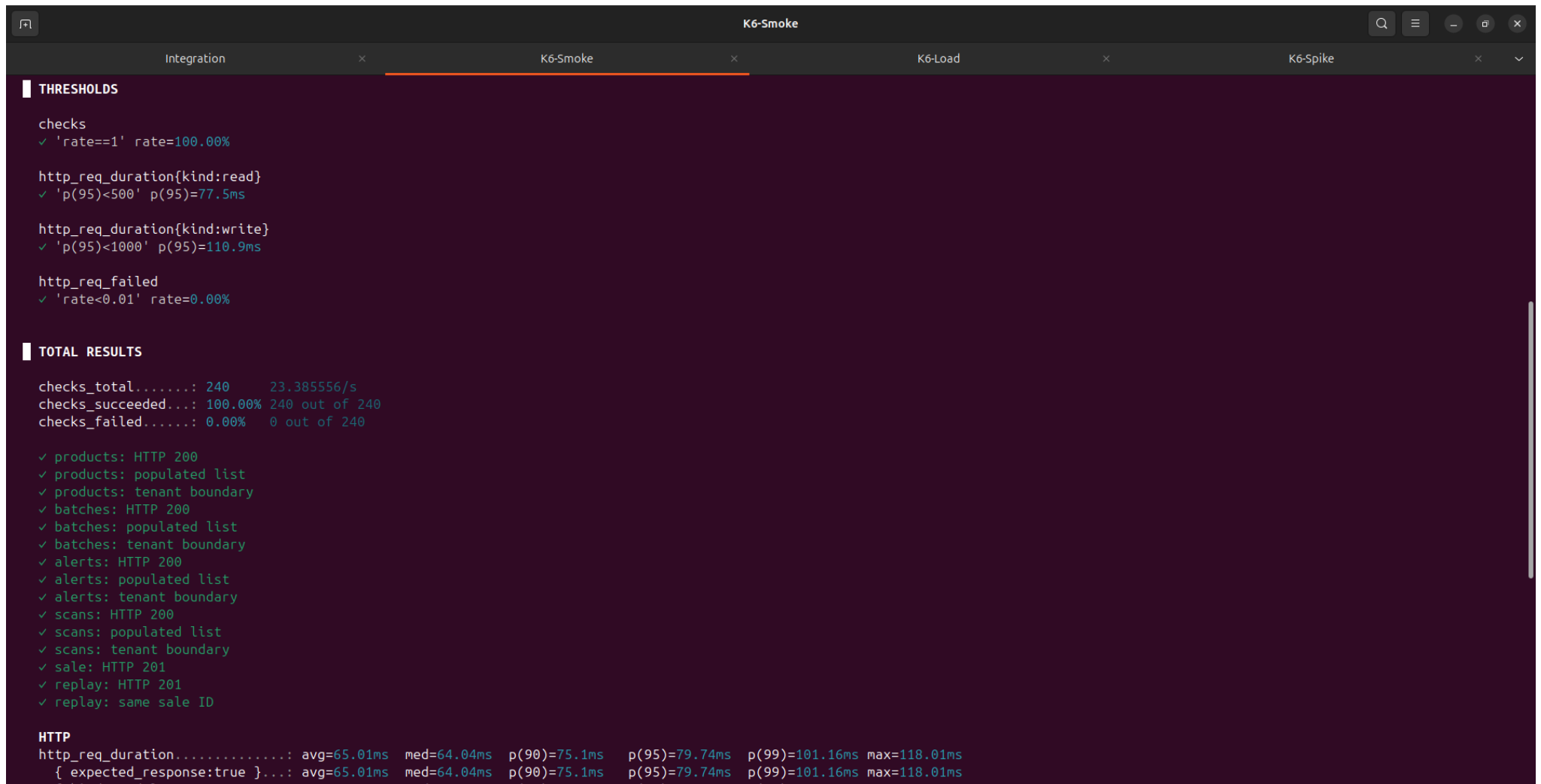
http_req_duration{kind:read}
✓ 'p(95)<500' p(95)=77.5ms

http_req_duration{kind:write}
```

E11. 1 virtual user for 10 seconds; local fixtures and thresholds shown. The gracefulStop warning caused the default 30-second grace period to apply.

Source: Screenshot from 2026-09-27 21-35-52.png · User-captured terminal screenshot

E12 — PERF-01: smoke thresholds and checks



```

K6-Smoke

Integration x K6-Smoke x K6-Load x K6-Spike x v

THRESHOLDS

checks
✓ 'rate==1' rate=100.00%

http_req_duration{kind:read}
✓ 'p(95)<500' p(95)=77.5ms

http_req_duration{kind:write}
✓ 'p(95)<1000' p(95)=110.9ms

http_req_failed
✓ 'rate<0.01' rate=0.00%

TOTAL RESULTS

checks_total.....: 240      23.385556/s
checks_succeeded...: 100.00% 240 out of 240
checks_failed.....: 0.00%  0 out of 240

✓ products: HTTP 200
✓ products: populated list
✓ products: tenant boundary
✓ batches: HTTP 200
✓ batches: populated list
✓ batches: tenant boundary
✓ alerts: HTTP 200
✓ alerts: populated list
✓ alerts: tenant boundary
✓ scans: HTTP 200
✓ scans: populated list
✓ scans: tenant boundary
✓ sale: HTTP 201
✓ replay: HTTP 201
✓ replay: same sale ID

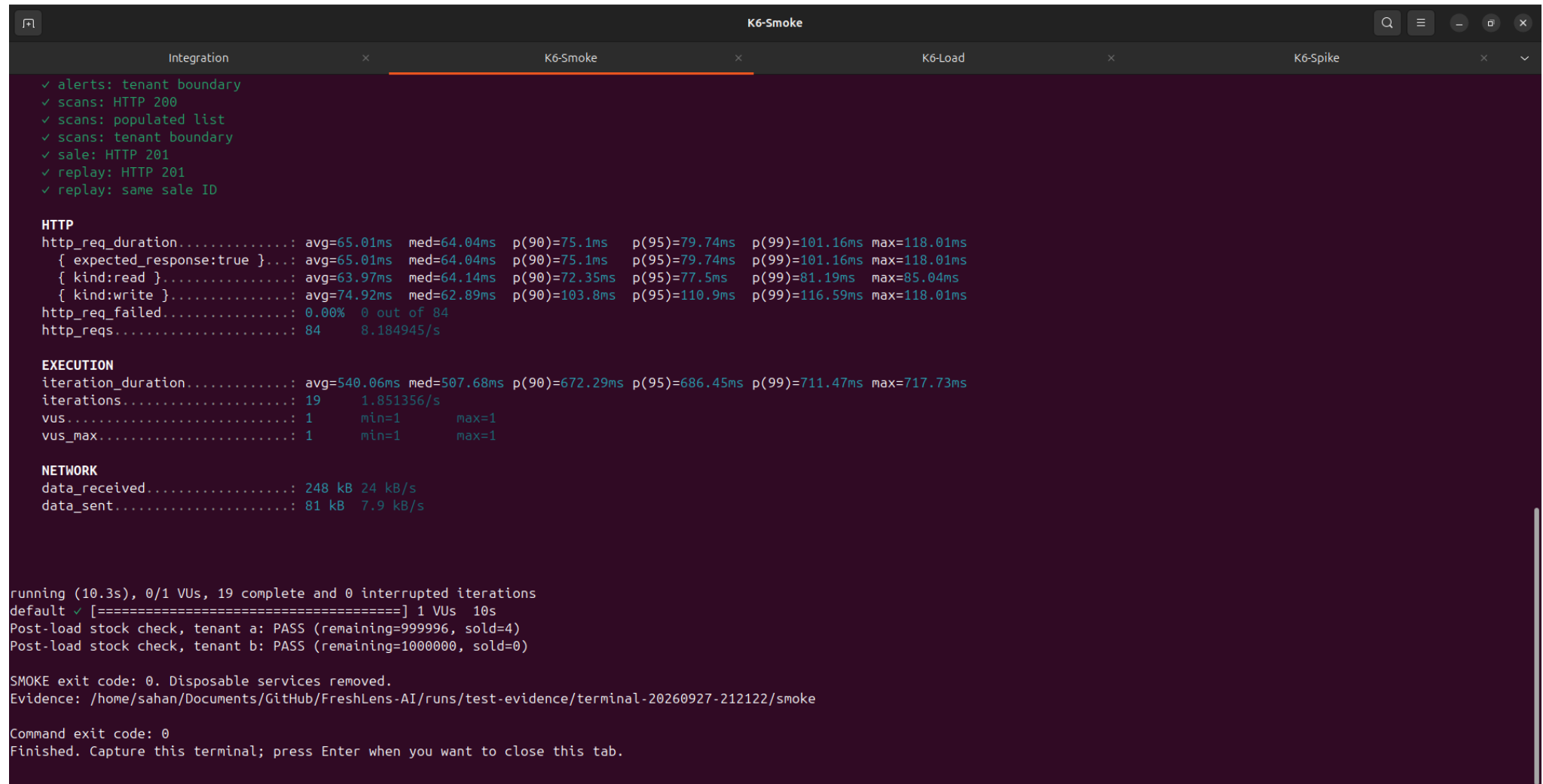
HTTP
http_req_duration.....: avg=65.01ms med=64.04ms p(90)=75.1ms p(95)=79.74ms p(99)=101.16ms max=118.01ms
{ expected_response:true }...: avg=65.01ms med=64.04ms p(90)=75.1ms p(95)=79.74ms p(99)=101.16ms max=118.01ms

```

E12. All four thresholds passed; 240 of 240 functional checks passed.

Source: Screenshot from 2026-09-27 21-35-57.png · User-captured terminal screenshot

E13 — PERF-01: smoke metrics and completion



```
Integration x K6-Smoke x K6-Load x K6-Spike x
✓ alerts: tenant boundary
✓ scans: HTTP 200
✓ scans: populated list
✓ scans: tenant boundary
✓ sale: HTTP 201
✓ replay: HTTP 201
✓ replay: same sale ID

HTTP
http_req_duration.....: avg=65.01ms med=64.04ms p(90)=75.1ms p(95)=79.74ms p(99)=101.16ms max=118.01ms
{ expected_response:true }...: avg=65.01ms med=64.04ms p(90)=75.1ms p(95)=79.74ms p(99)=101.16ms max=118.01ms
{ kind:read }.....: avg=63.97ms med=64.14ms p(90)=72.35ms p(95)=77.5ms p(99)=81.19ms max=85.04ms
{ kind:write }.....: avg=74.92ms med=62.89ms p(90)=103.8ms p(95)=110.9ms p(99)=116.59ms max=118.01ms
http_req_failed.....: 0.00% 0 out of 84
http_reqs.....: 84 8.184945/s

EXECUTION
iteration_duration.....: avg=540.06ms med=507.68ms p(90)=672.29ms p(95)=686.45ms p(99)=711.47ms max=717.73ms
iterations.....: 19 1.851356/s
vus.....: 1 min=1 max=1
vus_max.....: 1 min=1 max=1

NETWORK
data_received.....: 248 kB 24 kB/s
data_sent.....: 81 kB 7.9 kB/s

running (10.3s), 0/1 VUs, 19 complete and 0 interrupted iterations
default ✓ [=====] 1 VUs 10s
Post-load stock check, tenant a: PASS (remaining=999996, sold=4)
Post-load stock check, tenant b: PASS (remaining=1000000, sold=0)

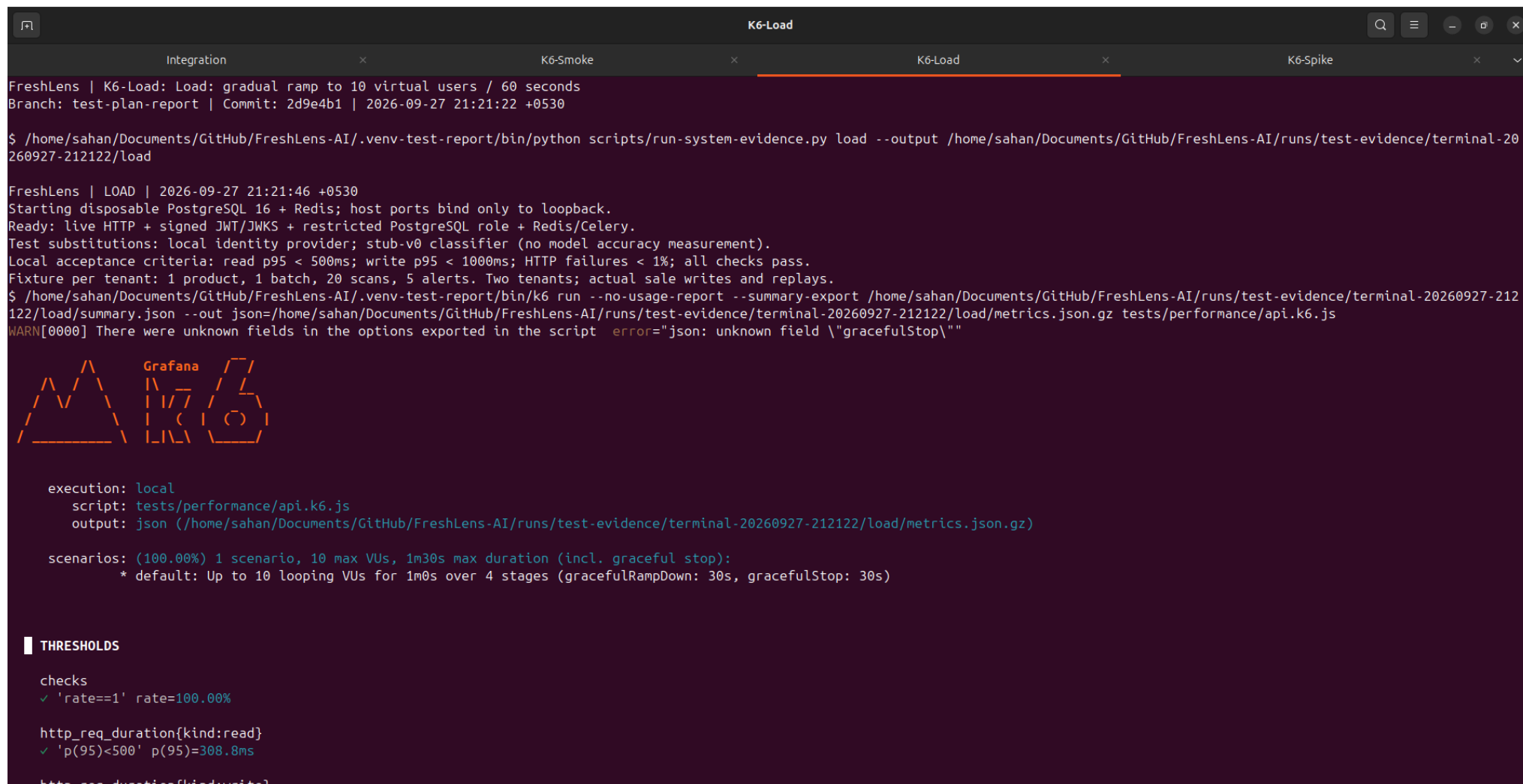
SMOKE exit code: 0. Disposable services removed.
Evidence: /home/sahan/Documents/GitHub/FreshLens-AI/runs/test-evidence/terminal-20260927-212122/smoke

Command exit code: 0
Finished. Capture this terminal; press Enter when you want to close this tab.
```

E13. 84 HTTP requests; read p95 77.50 ms, write p95 110.91 ms; 0% HTTP failures; both stock checks passed; exit 0.

Source: Screenshot from 2026-09-27 21-36-02.png · User-captured terminal screenshot

E14 — PERF-02: k6 load configuration



```
FreshLens | K6-Load: Load: gradual ramp to 10 virtual users / 60 seconds
Branch: test-plan-report | Commit: 2d9e4b1 | 2026-09-27 21:21:22 +0530

$ /home/sahan/Documents/GitHub/FreshLens-AI/.venv-test-report/bin/python scripts/run-system-evidence.py load --output /home/sahan/Documents/GitHub/FreshLens-AI/runs/test-evidence/terminal-20260927-212122/load

FreshLens | LOAD | 2026-09-27 21:21:46 +0530
Starting disposable PostgreSQL 16 + Redis; host ports bind only to loopback.
Ready: live HTTP + signed JWT/JWKS + restricted PostgreSQL role + Redis/Celery.
Test substitutions: local identity provider; stub-v0 classifier (no model accuracy measurement).
Local acceptance criteria: read p95 < 500ms; write p95 < 1000ms; HTTP failures < 1%; all checks pass.
Fixture per tenant: 1 product, 1 batch, 20 scans, 5 alerts. Two tenants; actual sale writes and replays.
$ /home/sahan/Documents/GitHub/FreshLens-AI/.venv-test-report/bin/k6 run --no-usage-report --summary-export /home/sahan/Documents/GitHub/FreshLens-AI/runs/test-evidence/terminal-20260927-212122/load/summary.json --out json=/home/sahan/Documents/GitHub/FreshLens-AI/runs/test-evidence/terminal-20260927-212122/load/metrics.json.gz tests/performance/api.k6.js
WARN[0000] There were unknown fields in the options exported in the script  error="json: unknown field \"gracefulStop\""



execution: local
  script: tests/performance/api.k6.js
  output: json (/home/sahan/Documents/GitHub/FreshLens-AI/runs/test-evidence/terminal-20260927-212122/load/metrics.json.gz)

scenarios: (100.00%) 1 scenario, 10 max VUs, 1m30s max duration (incl. graceful stop):
  * default: Up to 10 looping VUs for 1m0s over 4 stages (gracefulRampDown: 30s, gracefulStop: 30s)

THRESHOLDS

checks
✓ 'rate==1' rate=100.00%

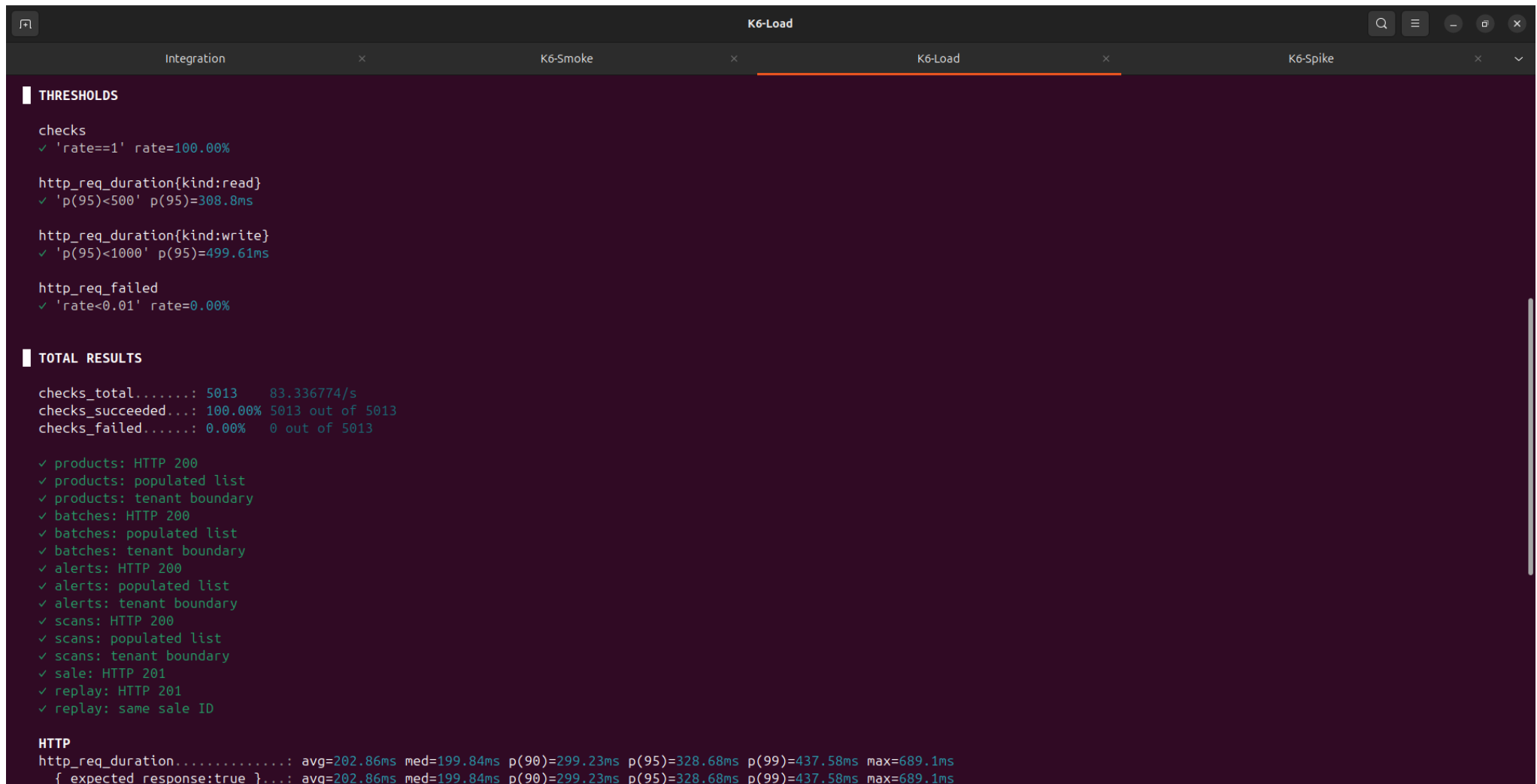
http_req_duration{kind:read}
✓ 'p(95)<500' p(95)=308.8ms

http_req_duration{kind:write}
```

E14. Gradual ramp to 10 virtual users over a 60-second profile; same populated fixtures and thresholds as smoke and spike.

Source: Screenshot from 2026-09-27 21-38-33.png · User-captured terminal screenshot

E15 — PERF-02: load thresholds and checks



```

K6-Load

Integration x K6-Smoke x K6-Load x K6-Spike v

THRESHOLDS

checks
✓ 'rate==1' rate=100.00%

http_req_duration{kind:read}
✓ 'p(95)<500' p(95)=308.8ms

http_req_duration{kind:write}
✓ 'p(95)<1000' p(95)=499.61ms

http_req_failed
✓ 'rate<0.01' rate=0.00%

TOTAL RESULTS

checks_total.....: 5013      83.336774/s
checks_succeeded...: 100.00% 5013 out of 5013
checks_failed.....: 0.00%   0 out of 5013

✓ products: HTTP 200
✓ products: populated list
✓ products: tenant boundary
✓ batches: HTTP 200
✓ batches: populated list
✓ batches: tenant boundary
✓ alerts: HTTP 200
✓ alerts: populated list
✓ alerts: tenant boundary
✓ scans: HTTP 200
✓ scans: populated list
✓ scans: tenant boundary
✓ sale: HTTP 201
✓ replay: HTTP 201
✓ replay: same sale ID

HTTP
http_req_duration.....: avg=202.86ms med=199.84ms p(90)=299.23ms p(95)=328.68ms p(99)=437.58ms max=689.1ms
{ expected_response:true }...: avg=202.86ms med=199.84ms p(90)=299.23ms p(95)=328.68ms p(99)=437.58ms max=689.1ms

```

E15. All four thresholds passed; 5,013 of 5,013 functional checks passed.

Source: Screenshot from 2026-09-27 21-38-39.png · User-captured terminal screenshot

E16 — PERF-02: load metrics and completion

```
Integration x K6-Smoke x K6-Load x K6-Spike v
✓ alerts: tenant boundary
✓ scans: HTTP 200
✓ scans: populated list
✓ scans: tenant boundary
✓ sale: HTTP 201
✓ replay: HTTP 201
✓ replay: same sale ID

HTTP
http_req_duration.....: avg=202.86ms med=199.84ms p(90)=299.23ms p(95)=328.68ms p(99)=437.58ms max=689.1ms
{ expected_response:true }...: avg=202.86ms med=199.84ms p(90)=299.23ms p(95)=328.68ms p(99)=437.58ms max=689.1ms
{ kind:read }.....: avg=196.03ms med=197.19ms p(90)=285.35ms p(95)=308.8ms p(99)=350.27ms max=416.83ms
{ kind:write }.....: avg=268.17ms med=252.83ms p(90)=445ms p(95)=499.61ms p(99)=639.39ms max=689.1ms
http_req_failed.....: 0.00% 0 out of 1754
http_reqs.....: 1754 29.158728/s

EXECUTION
iteration_duration.....: avg=1.14s med=1.1s p(90)=1.69s p(95)=1.78s p(99)=1.98s max=2.05s
iterations.....: 397 6.59978/s
vus.....: 1 min=1 max=10
vus_max.....: 10 min=10 max=10

NETWORK
data_received.....: 5.2 MB 86 kB/s
data_sent.....: 1.7 MB 28 kB/s

running (1m00.2s), 00/10 VUs, 397 complete and 0 interrupted iterations
default ✓ [=====] 00/10 VUs 1m0s
Post-load stock check, tenant a: PASS (remaining=999958, sold=42)
Post-load stock check, tenant b: PASS (remaining=999959, sold=41)

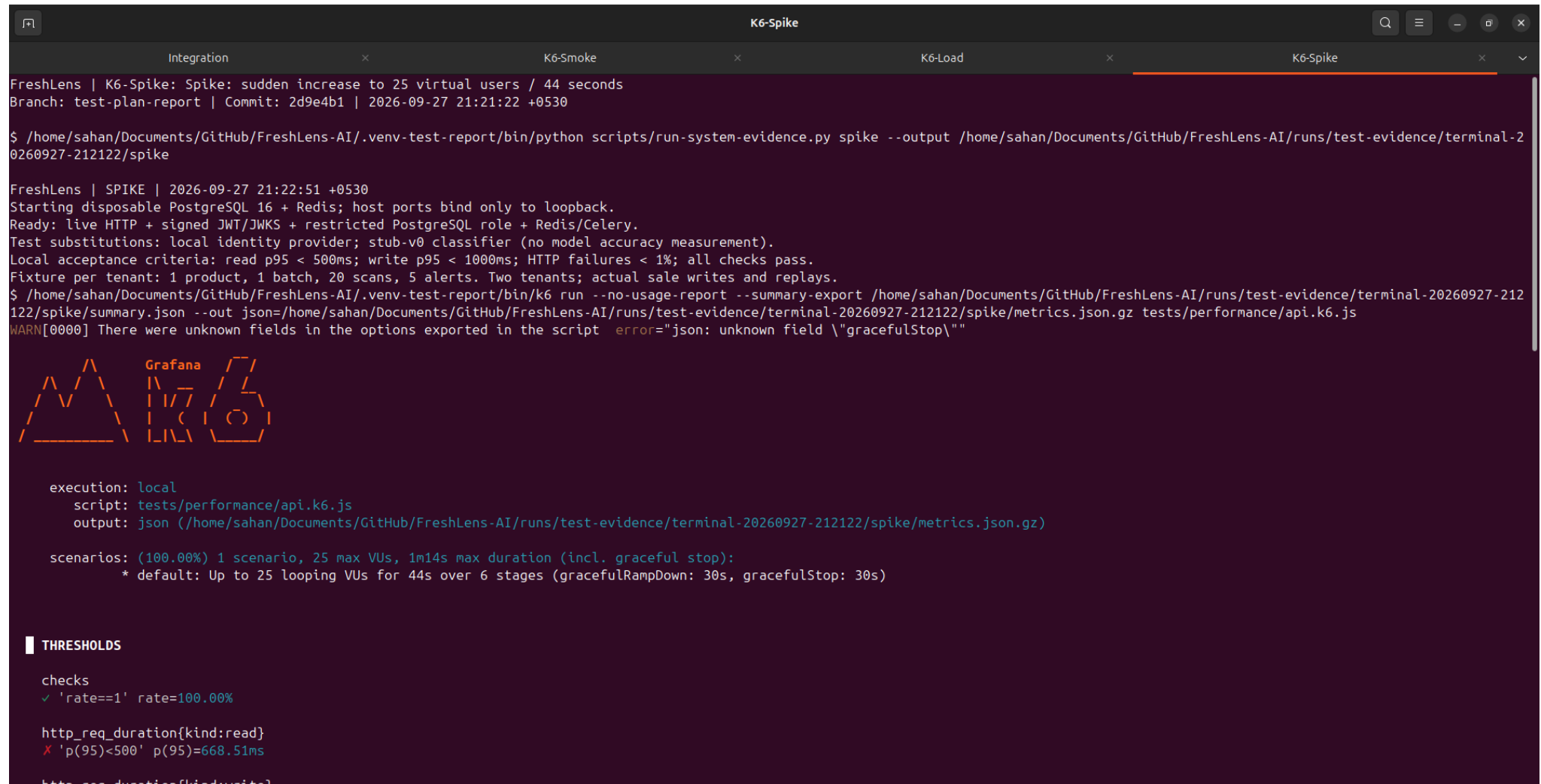
LOAD exit code: 0. Disposable services removed.
Evidence: /home/sahan/Documents/GitHub/FreshLens-AI/runs/test-evidence/terminal-20260927-212122/load

Command exit code: 0
Finished. Capture this terminal; press Enter when you want to close this tab.
```

E16. 1,754 HTTP requests; read p95 308.80 ms, write p95 499.62 ms (rounded from JSON); 0% HTTP failures; stock checks passed; exit 0.

Source: Screenshot from 2026-09-27 21-38-41.png · User-captured terminal screenshot

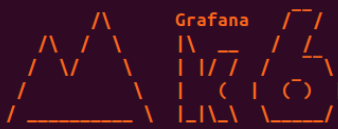
E17 — PERF-03: k6 spike configuration



```
FreshLens | K6-Spike: Spike: sudden increase to 25 virtual users / 44 seconds
Branch: test-plan-report | Commit: 2d9e4b1 | 2026-09-27 21:21:22 +0530

$ /home/sahan/Documents/GitHub/FreshLens-AI/.venv-test-report/bin/python scripts/run-system-evidence.py spike --output /home/sahan/Documents/GitHub/FreshLens-AI/runs/test-evidence/terminal-20260927-212122/spike

FreshLens | SPIKE | 2026-09-27 21:22:51 +0530
Starting disposable PostgreSQL 16 + Redis; host ports bind only to loopback.
Ready: live HTTP + signed JWT/JWKS + restricted PostgreSQL role + Redis/Celery.
Test substitutions: local identity provider; stub-v0 classifier (no model accuracy measurement).
Local acceptance criteria: read p95 < 500ms; write p95 < 1000ms; HTTP failures < 1%; all checks pass.
Fixture per tenant: 1 product, 1 batch, 20 scans, 5 alerts. Two tenants; actual sale writes and replays.
$ /home/sahan/Documents/GitHub/FreshLens-AI/.venv-test-report/bin/k6 run --no-usage-report --summary-export /home/sahan/Documents/GitHub/FreshLens-AI/runs/test-evidence/terminal-20260927-212122/spike/summary.json --out json=/home/sahan/Documents/GitHub/FreshLens-AI/runs/test-evidence/terminal-20260927-212122/spike/metrics.json.gz tests/performance/api.k6.js
WARN[0000] There were unknown fields in the options exported in the script  error="json: unknown field \"gracefulStop\""
```



```
execution: local
  script: tests/performance/api.k6.js
  output: json (/home/sahan/Documents/GitHub/FreshLens-AI/runs/test-evidence/terminal-20260927-212122/spike/metrics.json.gz)

scenarios: (100.00%) 1 scenario, 25 max VUs, 1m14s max duration (incl. graceful stop):
  * default: Up to 25 looping VUs for 44s over 6 stages (gracefulRampDown: 30s, gracefulStop: 30s)

THRESHOLDS

checks
✓ 'rate==1' rate=100.00%

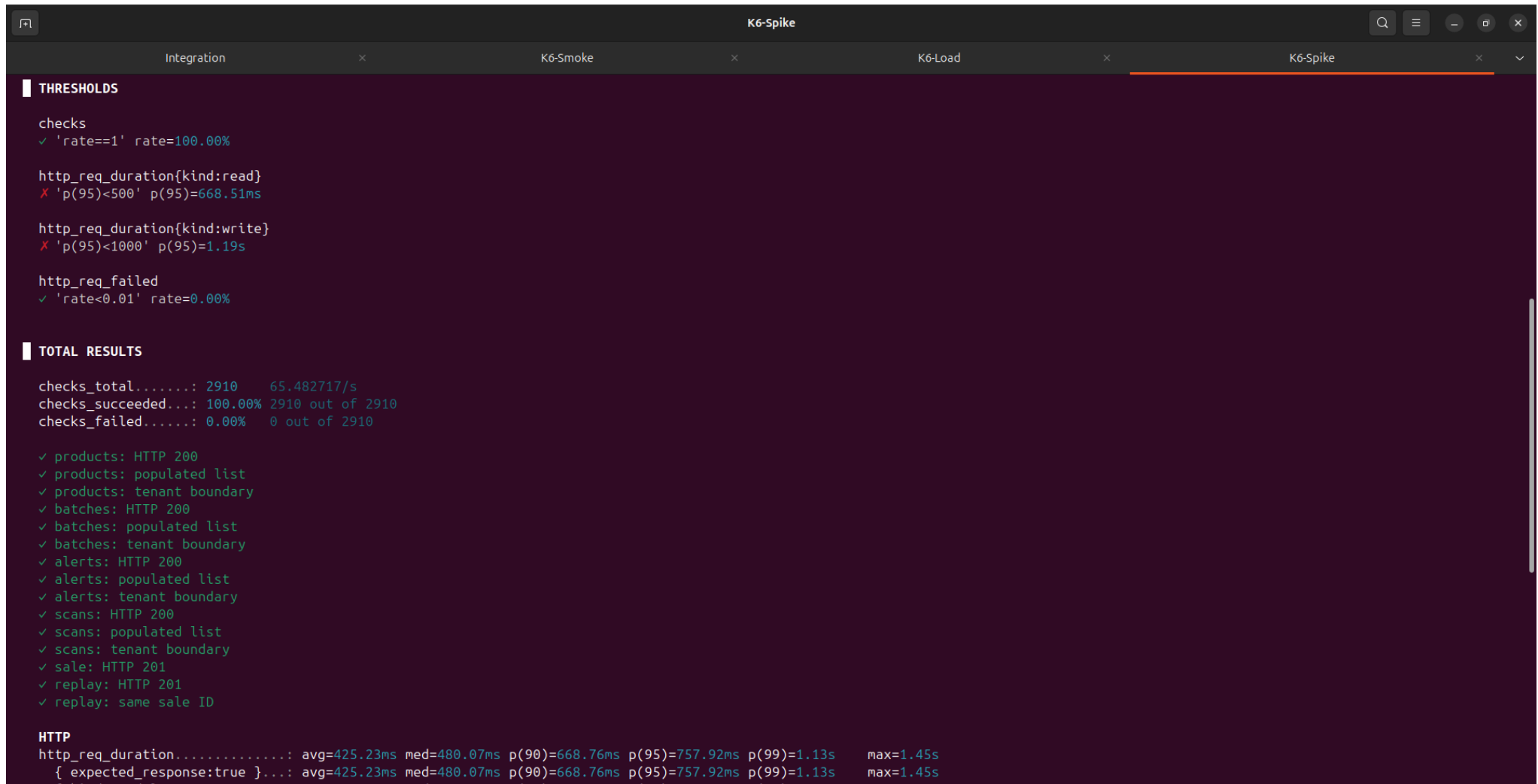
http_req_duration{kind:read}
✗ 'p(95)<500' p(95)=668.51ms

http_req_duration{kind:write}
```

E17. Short spike to 25 virtual users over a 44-second profile; setup, actual grace period and acceptance thresholds shown.

Source: Screenshot from 2026-09-27 21-38-46.png · User-captured terminal screenshot

E18 — PERF-03: spike latency threshold failures



```
THRESHOLDS

checks
✓ 'rate==1' rate=100.00%

http_req_duration{kind:read}
✗ 'p(95)<500' p(95)=668.51ms

http_req_duration{kind:write}
✗ 'p(95)<1000' p(95)=1.19s

http_req_failed
✓ 'rate<0.01' rate=0.00%

TOTAL RESULTS

checks_total.....: 2910      65.482717/s
checks_succeeded...: 100.00% 2910 out of 2910
checks_failed.....: 0.00%   0 out of 2910

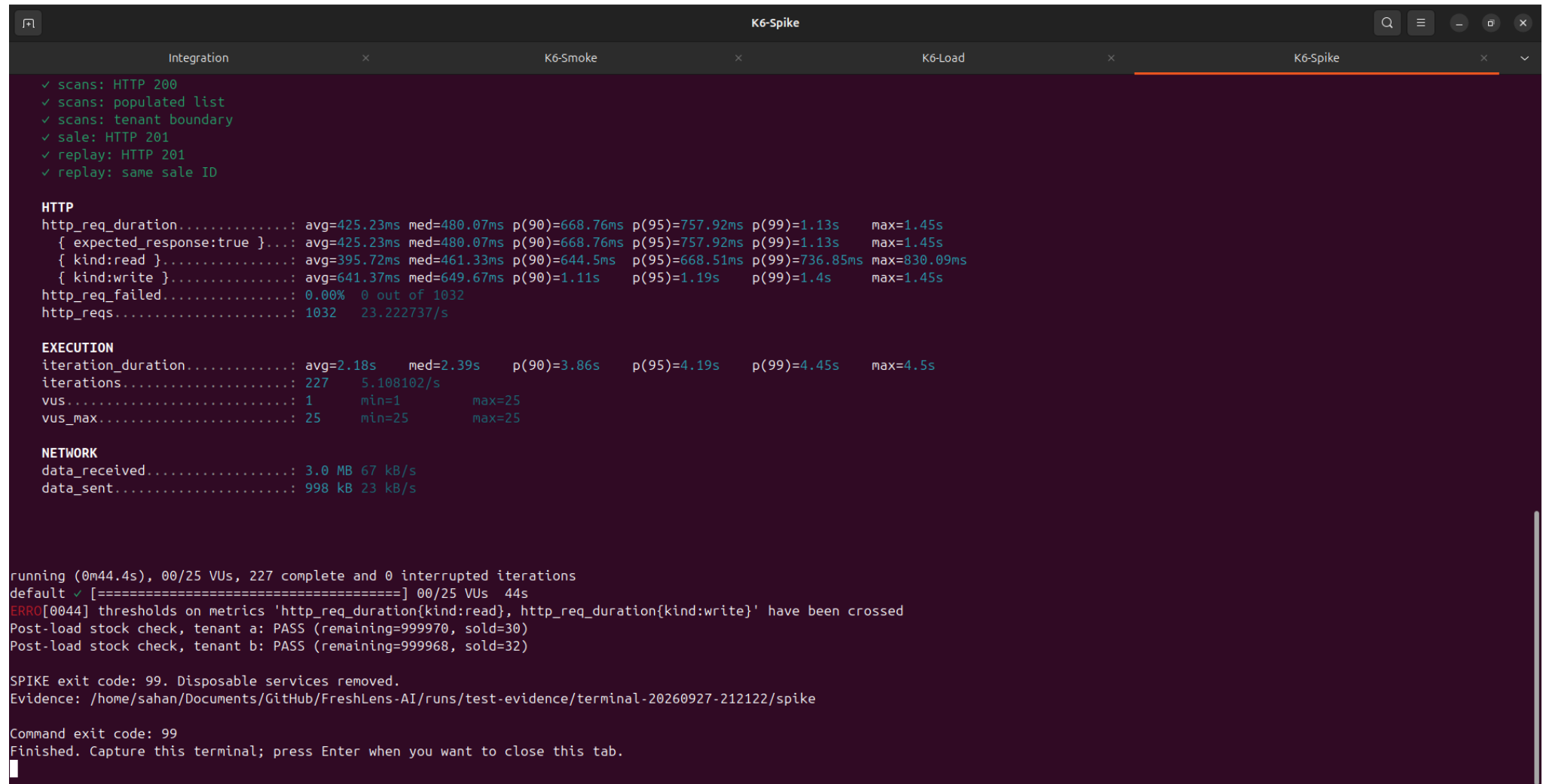
✓ products: HTTP 200
✓ products: populated list
✓ products: tenant boundary
✓ batches: HTTP 200
✓ batches: populated list
✓ batches: tenant boundary
✓ alerts: HTTP 200
✓ alerts: populated list
✓ alerts: tenant boundary
✓ scans: HTTP 200
✓ scans: populated list
✓ scans: tenant boundary
✓ sale: HTTP 201
✓ replay: HTTP 201
✓ replay: same sale ID

HTTP
http_req_duration.....: avg=425.23ms med=480.07ms p(90)=668.76ms p(95)=757.92ms p(99)=1.13s max=1.45s
{ expected_response:true }...: avg=425.23ms med=480.07ms p(90)=668.76ms p(95)=757.92ms p(99)=1.13s max=1.45s
```

E18. Read p95 668.52 ms exceeds 500 ms; write p95 1,192.94 ms exceeds 1,000 ms. All 2,910 functional checks passed, but the performance verdict is FAIL.

Source: Screenshot from 2026-09-27 21-38-50.png · User-captured terminal screenshot

E19 — PERF-03: spike metrics and failed completion



```
Integration x K6-Smoke x K6-Load x K6-Spike x
✓ scans: HTTP 200
✓ scans: populated list
✓ scans: tenant boundary
✓ sale: HTTP 201
✓ replay: HTTP 201
✓ replay: same sale ID

HTTP
http_req_duration.....: avg=425.23ms med=480.07ms p(90)=668.76ms p(95)=757.92ms p(99)=1.13s max=1.45s
  { expected_response:true }...: avg=425.23ms med=480.07ms p(90)=668.76ms p(95)=757.92ms p(99)=1.13s max=1.45s
  { kind:read }.....: avg=395.72ms med=461.33ms p(90)=644.5ms p(95)=668.51ms p(99)=736.85ms max=830.09ms
  { kind:write }.....: avg=641.37ms med=649.67ms p(90)=1.11s p(95)=1.19s p(99)=1.4s max=1.45s
http_req_failed.....: 0.00% 0 out of 1032
http_reqs.....: 1032 23.222737/s

EXECUTION
iteration_duration.....: avg=2.18s med=2.39s p(90)=3.86s p(95)=4.19s p(99)=4.45s max=4.5s
iterations.....: 227 5.108102/s
vus.....: 1 min=1 max=25
vus_max.....: 25 min=25 max=25

NETWORK
data_received.....: 3.0 MB 67 kB/s
data_sent.....: 998 kB 23 kB/s

running (0m44.4s), 00/25 VUs, 227 complete and 0 interrupted iterations
default ✓ [=====] 00/25 VUs 44s
ERRO[0044] thresholds on metrics 'http_req_duration{kind:read}, http_req_duration{kind:write}' have been crossed
Post-load stock check, tenant a: PASS (remaining=999970, sold=30)
Post-load stock check, tenant b: PASS (remaining=999968, sold=32)

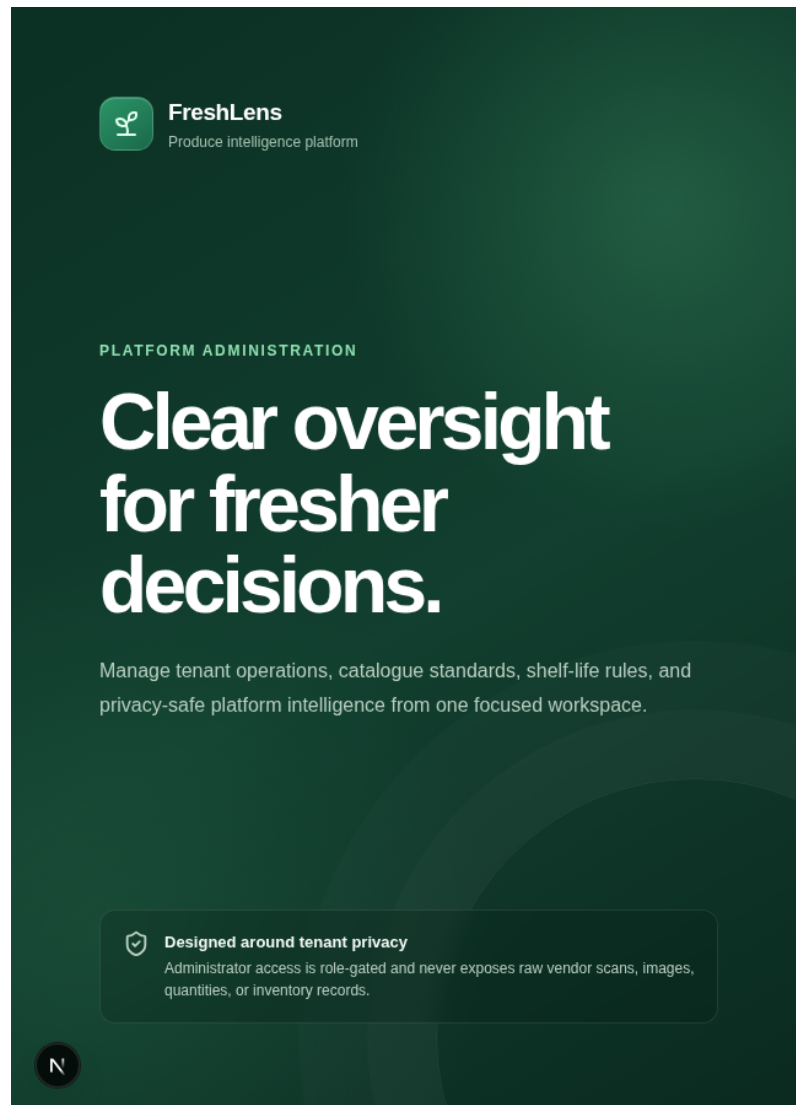
SPIKE exit code: 99. Disposable services removed.
Evidence: /home/sahan/Documents/GitHub/FreshLens-AI/runs/test-evidence/terminal-20260927-212122/spike

Command exit code: 99
Finished. Capture this terminal; press Enter when you want to close this tab.
```

E19. 1,032 HTTP requests; 0% HTTP failures; stock checks passed for both tenants. Exit code 99 records the two failed latency thresholds.

Source: Screenshot from 2026-09-27 21-38-53.png · User-captured terminal screenshot

E20 — UI-02: Playwright desktop form validation



RESTRICTED ACCESS

Welcome back

Sign in with the administrator account provisioned by your FreshLens project owner.

Email address

admin@freshlens.ai

Enter a valid administrator email.

Password

Supabase protected

Enter your password

Enter your password.

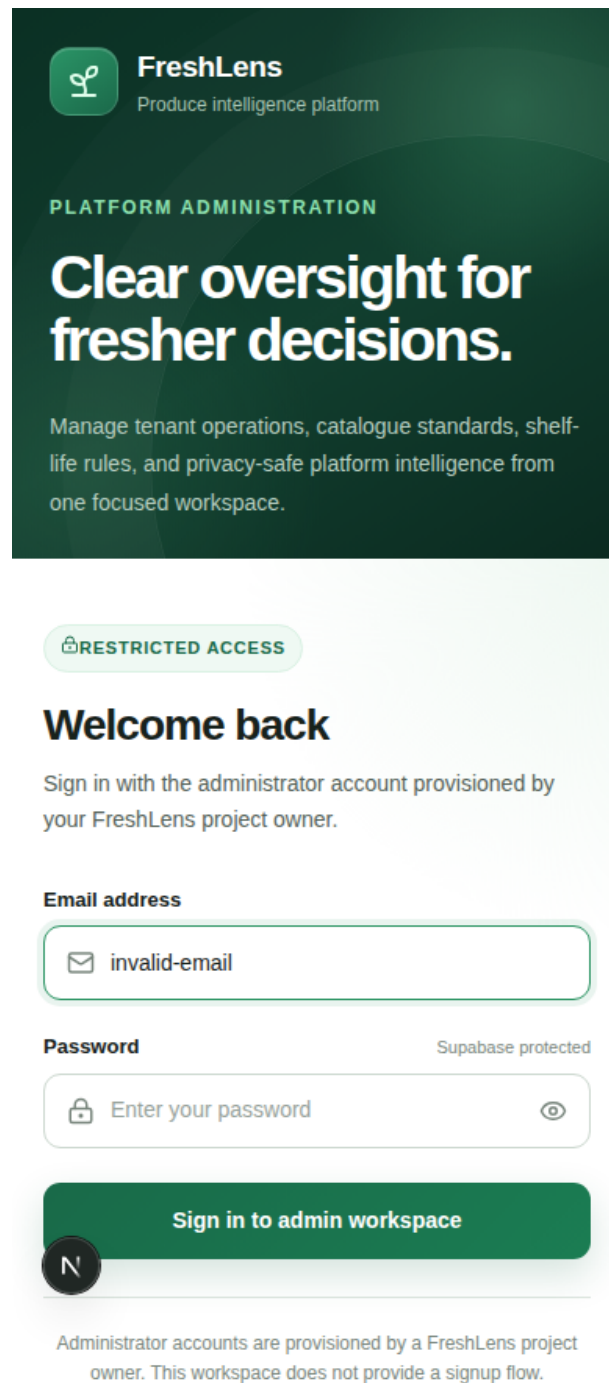
Sign in to admin workspace

Administrator accounts are provisioned by a FreshLens project owner. This workspace does not provide a signup flow.

E20. Actual Next.js sign-in page after Playwright submitted an empty form. Both validation messages and invalid-field states were asserted at 1440 x 900.

Source: test_empty_submission_shows_field_errors.png · Playwright browser screenshot, 27 September 2026

E21 — UI-02: Playwright mobile-width layout



The image shows a mobile-width login screen for the FreshLens platform. The top section has a dark green background with the FreshLens logo and tagline 'Produce intelligence platform'. Below this, the text 'PLATFORM ADMINISTRATION' is displayed in a light green font. The main heading is 'Clear oversight for fresher decisions.' in white. A subtext describes the platform's capabilities: 'Manage tenant operations, catalogue standards, shelf-life rules, and privacy-safe platform intelligence from one focused workspace.' A light green button with a lock icon and the text 'RESTRICTED ACCESS' is positioned above the login form. The form itself is on a light green background and includes a 'Welcome back' message, a sign-in instruction, and two input fields: 'Email address' (containing 'invalid-email') and 'Password' (containing 'Enter your password'). A green 'Sign in to admin workspace' button is at the bottom of the form. A small circular icon with the letter 'N' is located to the left of the button. At the very bottom, a note states: 'Administrator accounts are provisioned by a FreshLens project owner. This workspace does not provide a signup flow.'

FreshLens
Produce intelligence platform

PLATFORM ADMINISTRATION

Clear oversight for fresher decisions.

Manage tenant operations, catalogue standards, shelf-life rules, and privacy-safe platform intelligence from one focused workspace.

RESTRICTED ACCESS

Welcome back

Sign in with the administrator account provisioned by your FreshLens project owner.

Email address

 invalid-email

Password Supabase protected

 Enter your password 

Sign in to admin workspace

N

Administrator accounts are provisioned by a FreshLens project owner. This workspace does not provide a signup flow.

E21. Actual browser capture at a 390 x 844 viewport (full page). Playwright verified the submit control was visible and inside the viewport width, with no horizontal overflow. This is browser viewport testing, not native mobile testing.

Source: test_mobile_width_login_has_no_horizontal_overflow.png · Playwright browser screenshot, 27 September 2026